

When Does Trace-Driven Evaluation Mislead MoE Expert Caching?

Replay Semantics, Workload Contamination, and Operating Regimes

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Abstract

Mixture-of-Experts (MoE) models have outgrown the high-bandwidth memory of the accelerators that serve them, and offloading expert weights to host memory has become a standard response. This makes expert cache management an attractive lever: if a smarter policy raised the hit rate, the same model would need less expert traffic per token. Evaluating that hypothesis is a measurement problem, and we find the measurement fragile.

Using a trace-driven, event-atomic simulator over three MoE models (40, 64 and 128 experts), we isolate three evaluation axes that change conclusions rather than shift numbers. *Replay semantics*: under a fused-event traffic contract, an inconsistent per-access replay inflates recency-based policies by 27–29% while leaving frequency-based and static policies within 4%, inverting the policy ranking. *Workload contamination*: probe sets using one instruction template per category produce verbatim-identical generation prefixes; a matched-pair rendering intervention moves the measured early-window effect by 19.4–31.9 percentage points and reverses which workloads appear most cache-friendly. *Operating regimes*: normalized miss fractions do not transfer across models, so the per-step expert union relative to per-layer capacity must be reported — though permuting only the temporal order of an otherwise identical event stream moves the offline-optimal gap from 44.9% to 30.8%, so it is not sufficient.

After correcting all three, a stable gap to the offline optimum remains (44.2–45.9% across 13 frozen workload compositions). A forced-admission oracle attributes 84.3–96.6% of it to knowing which resident expert is used furthest in the future. A causal next-use predictor, used as an eviction rule, recovers –11.4% of the gap; at the decision points it selects an optimal victim 3.4% of the time against 2.4% for a random resident block and 20.6–22.1% for LRU and LFRU. Our position is narrow: **in our evaluated settings a large offline-optimal gap substantially overstates the gains recovered by representative lightweight causal mechanisms**. On publication we will release the simulator, a diversity-controlled probe set, the contamination diagnostics and a reporting checklist, subject to the upstream licences recorded in the artifact matrix.

1. Introduction

Serving a frontier Mixture-of-Experts model no longer fits the memory it was designed for. A contemporary large MoE dedicates on the order of 1.3 TiB to routed expert weights while activating only a small fraction per token, so recent systems keep a subset of experts resident in device memory and stream the rest from host memory on demand. Variants appear in open-source prototypes, research systems, and inference-engine proposals; their execution contracts and performance targets differ (Shazeer et al. 2017; Fedus et al. 2022; Eliseev and Mazur 2023; Xue et al. 2025; Tang et al. 2024).

The design also creates an obvious hypothesis. If experts are cached, then a better cache policy should raise the hit rate, lower host-to-device traffic, and let the same model run on fewer accelerators. A workload-aware controller — one that learns which experts a customer’s traffic actually touches — would then be worth building. A substantial body of recent work pursues variants of this idea through prefetching, learned routing prediction, and specialized replacement policies (Du et al. 2024; Fang et al. 2025; Hoang et al. 2026; Zhu et al. 2026).

Whether the hypothesis holds is an empirical question, and answering it requires measuring how much room a policy has: the distance between the best causal policy and an offline optimum. We set out to measure exactly that. What we found first was that the measurement itself is unstable in ways that are easy to miss and that change the answer, not just its precision.

Three axes. Within our trace-driven study, we identify three evaluation choices, each individually plausible and each capable of reversing a conclusion. Their applicability is not identical: Axis I concerns replay of a claimed fused-event contract and does not apply retrospectively to a real end-to-end system whose implementation defines its own transfer semantics (§12); Axes II and III concern workload construction and operating-regime reporting more broadly.

Replay semantics. Under the fused-event traffic contract studied here, the experts a scheduling step touches at a layer form one committed execution unit. A simulator that flattens this unit into individual accesses can evict, part-way through the event, an expert that was resident at event start and is required later, then count an artificial refetch. The distortion is not uniform noise. On Qwen3-30B-A3B at $\rho=40\%$ and $B=8$, sequential replay inflates LRU by 27.4%, LFRU by 28.5%, and Least-Stale by 27.4%, but LFU by only 3.4%, Belady by 4.3%, and a same-trace static diagnostic not at all. Because it penalizes exactly one family of policies, it inverts their ranking: under sequential replay the same-trace static diagnostic appears to beat every dynamic policy; under event-atomic replay LFRU beats it by 5.7%. This is a replay-invariance comparison, not a deployable static baseline (§2.3). In our own earlier measurements, correcting this semantics moved the steady-state prefill gap on two models from a range that looked like algorithmic headroom down to 0.155–3.169%.

Workload contamination. Studies of workload-conditioned routing need probe sets partitioned by task. The natural construction — one instruction template per category, filled with different payloads — turns out to be hazardous. Under raw continuation, or within a short decode window, the model reproduces the shared template before it produces anything task-specific, so concurrent requests emit verbatim-identical prefixes at identical decode positions. In one of our own probe categories, 4 of 16 requests generated byte-identical 63-token outputs. The resulting expert overlap is easily mistaken for semantic locality: restricting the comparison to request pairs whose generated text barely overlaps collapses one category’s within-class expert Jaccard from 0.229 to 0.092, against a cross-category baseline of 0.078. We introduce a matched-pair design — the same source records rendered both with diverse and with fixed templates — that changes the early-window effect by 19.4–31.9 percentage points. An abrupt decay aligned with a shared generated prefix is a diagnostic of surface repetition, though genuine task effects may also vary by response phase. Correcting the construction reverses the point-estimate ordering; only two of six corrected category effects replicate clearly across two request draws.

Operating regimes. Normalized miss fractions are routinely compared across MoE models with different expert counts. They should not be. The relevant variable is the ratio of the per-step, per-layer expert union to the per-layer cache capacity. As the fraction of events exceeding capacity grows, capacity-forced traffic increases and policy headroom can collapse — so a small gap can mean “the capacity floor dominates” rather than “existing policies suffice”. Holding the model and the workload fixed and changing only the cache fraction moves the gap from 7.4% to 33.3%. The ratio is necessary but not sufficient: permuting only the temporal order of an event stream, with the event set and the ratio held exactly constant, moves the gap from 44.9% to 30.8% and changes which causal policy is best.

What remains after correction. With all three controlled, a large gap to the offline optimum persists: 44.2–45.9% across 13 frozen workload compositions at $\rho=40\%$ and $B=8$, and 50.4% at $B=2$. The natural reading is that a substantial opportunity is waiting for a better online policy. We tested that reading directly.

Belady’s advantage over a causal policy has two sources: refusing to admit a block that will not be reused before it would be evicted, and knowing which resident block is used furthest in the future. Only the first has a clear online analogue. Replacing the oracle with one that is forced to admit every miss separates them: bypass admission accounts for 15.7% of the gap at $B=8$ and 3.4% at $B=2$, leaving 84.3% and 96.6% to future-victim knowledge. We then trained a next-use-distance predictor on causal features available at eviction time — recency, frequency, gate mass, concurrent routing multiplicity, layer, and popularity rate — fitted on one trace and evaluated on a disjoint one. It transfers at $R^2=0.24$ and, substituted for the oracle’s input in the same eviction and admission machinery, recovers -11.4% of the gap: it is worse than the causal baseline it was meant to improve.

Position. We state our claim narrowly. We do not show that expert caching is a closed problem, that future-victim knowledge is unpredictable in principle, or that online MoE cache controllers have no future; existing work on expert *prefetching* predicts a different quantity — which experts will be routed next, rather than how long a cached block will remain unused — and is not contradicted by our result. What we show is that in our evaluated settings a large offline-optimal gap substantially overstates the gains recovered by representative lightweight causal mechanisms, and that reporting such a gap without decomposing it invites an unwarranted inference. We accompany this with three scoped negative results and one invalidated mechanism experiment — steady-state prefill eviction, warm-cache workload transition, semantic cache partitioning, and causal affinity batching — each annotated with the model, cache fraction, concurrency, scheduling discipline, and waiting constraint under which it was measured, because we found that at least one of these mechanisms fails for a reason that does not extrapolate.

Contributions.

1. We formalize and implement an **event-atomic replay protocol** for the fused-event traffic contract used in our trace-driven evaluation, show that an inconsistent flattened replay selectively penalizes recency-based policies and inverts policy rankings, and provide a hand-verifiable reference trace (§2, §4).
2. We introduce a **matched-pair method** for quantifying prompt-template contamination in workload-conditioned routing studies, together with prompt-side and route-side diagnostics, and release a diversity-controlled probe set built to be free of it (§5).
3. We show that **cross-model comparison of MoE cache results requires explicitly controlling the operating regime**, give the union-to-capacity ratio as a first-order variable that must be reported, and demonstrate by ablation that it is necessary but not sufficient (§6).
4. We **decompose the offline-optimal gap** into an online-approximable and a future-dependent component and show, for representative causal mechanisms in our settings, a large gap between the oracle bound and realizable gains (§7, §8).
5. We distill the above into a **reporting checklist** for future MoE cache evaluations (§10) and will release, on publication, the simulator, probe set, contamination tools, frozen manifests and — where upstream licences permit — routing artifacts and pre-registration documents including failed criteria (§13).

Scope. All results are trace-driven simulation; we do not measure real host-to-device transfers, kernel time, or interconnect topology. Our primary results use a single model, with two smaller models for cross-model checks, and we evaluate only strictly lossless policies — no expert substitution, pruning, or reduced-precision fallback. Section 11 states these limits in full. We regard the checklist in §10, rather than any individual number, as the most portable result of this work.

Relation to prior evaluations. We audited the public reporting of ten representative papers (§12). Seven are end-to-end systems, so our replay failure is not applicable to their actual execution; one is trace-driven but does not report enough detail to reconstruct our fused-event contract and assumes batch size one; two study architecture or prediction targets. None directly reports the measured union-to-capacity ratio needed to place its results in §6’s regime, and prompt-template multiplicity is generally not reported. These omissions do not make the results incorrect. They prevent direct comparison under a common evaluation contract, which is the motivation for §10’s checklist.

§2 Evaluation Model

Results in this area are difficult to compare because the underlying replay model is rarely stated. We therefore fix notation and semantics before reporting any number. Everything in §4–§8 is defined against this model, and a hand-verifiable reference trace is released with the artifacts.

2.1 Traces and events

A model has L MoE layers and N routed experts per layer, and routes each token to k experts per layer. The unit of cache management is an **expert block** $\mathbf{b} = (l, \mathbf{e})$ with l in $[L]$, $\mathbf{e}$ in $[N]$; the block universe has size $L \cdot N$.

Requests are served under first-come-first-served continuous batching. At scheduling step $\mathbf{s}$, $A(\mathbf{s})$ denotes the set of active requests. For the fused-event traffic contract evaluated in this paper, the quantity that matters for memory traffic is not the per-token expert assignment but its **union over the concurrent batch**:

Definition 1 (event). An *event* is a pair $(\mathbf{s}, l)$. Its expert set is $E(\mathbf{s}, l) = \text{union}_{\mathbf{r} \in A(\mathbf{s})} \text{topk}(\mathbf{r}, l, \mathbf{s})$, the distinct expert blocks that step $\mathbf{s}$ touches at layer l .

A **trace** is the sequence of events ordered lexicographically by $(\mathbf{s}, l)$. Deduplication inside an event is a property of the execution unit, not of any cache policy: an expert selected by five concurrent requests is fetched once. We therefore treat it as free and never credit it to a policy (§2.4).

2.2 Event-atomic replay

Definition 2 (event-atomic replay). Let C be the cache contents. For each event $(\mathbf{s}, l)$ in trace order:

1. take the snapshot C at event start;
2. classify every member of $E(\mathbf{s}, l)$ against that snapshot: $\text{hits} = E \text{ intersect } C, \text{misses} = E \setminus C$;
3. serve **all** of **misses** (no partial service);
4. only after the event completes, apply admission and eviction to obtain C' .

The ordering constraint is step 4. Admission and eviction are deferred to the event boundary, so no member of $E \text{ intersect } C$ can be evicted before the event that needs it has completed. This counts each missing expert once and assumes the execution workspace can stream that expert, compute all tokens routed to it, and release temporary state before choosing the retained cache contents. It does **not** assume that all members of E fit in the cache simultaneously.

Sequential replay, the alternative we examine in §4, flattens $E(\mathbf{s}, l)$ into an ordered list of individual accesses and applies admission and eviction after each one. When $|E(\mathbf{s}, l)|$ exceeds the layer’s quota, an access late in the event can evict a block in $E \text{ intersect } C$ that the *same* event has not yet consumed, which is then counted as a miss.

A remark on modelling. We do not claim event-atomic replay is the only admissible model of MoE execution. An engine that serializes expert loads within a layer is a legitimate alternative, and a simulator may reasonably model it. The distinction we draw is narrower and is a correctness property, not a preference:

Sequential *execution* is a realizable system. What is inconsistent is to claim one-transfer-per-distinct-expert fused-event accounting while evicting a block resident at event start, still required later in that committed event, and counting its refetch as if it were a new demand.

A simulator should state which contract it targets. Section 4 measures what happens when a flattened replay is used to estimate the fused-event contract without protecting start-resident, not-yet-served members.

2.3 Cache scope, capacity and policies

Capacity is a residency fraction ρ of the block universe; the resident set at any instant is the analogue of a working set in the classical sense (Denning 1968). Under **per-layer scope** the budget is divided into quotas c_l , with the remainder to the lowest-indexed layers; under **global scope** a single pool serves all layers. Per-layer scope is primary throughout, because expert weights are consumed layer-by-layer within a step.

We evaluate **Demand** (no cache), **Static** (a fixed set chosen offline from a calibration trace and never updated), **LRU**, **LFU**, **LFRU** (rank by frequency / age), **Least-Stale** (blocks untouched in the current forward cycle are evicted first), **Belady** (Belady 1966) (offline optimum with bypass admission), and **Belady-forced-admit**, which is not deployable but is the decomposition probe of §7.2. Ties break by seeded random priority; §7.1 reports tie-seed sensitivity. Two static pin protocols appear in this paper and are named where used: a **calibration-fitted** list, frozen from a disjoint trace and used in §9, and a **same-trace diagnostic** list used only in Table 2, where the question is whether replay semantics moves a policy that by construction cannot move. The initial load of a pinned set is counted in transferred blocks and reported separately from steady-state traffic.

2.4 Metrics

Let M be the number of blocks transferred over a trace, and define

- **logical** — expert assignments **before** intra-event deduplication. For the decode traces used in the primary experiments this is $\sum_{\{s,l\}} |A(s,l)| \cdot k$; for prefill it must additionally sum the routed tokens contributed by each request rather than merely the request count;
- **union** = $\sum_{\{s,l\}} |E(s,l)|$ — distinct blocks touched.

We report

$$\text{effective miss fraction } m = M / \text{logical} \quad \text{recoverable gap } g = (m_{\text{best-causal}} - m_{\text{Belady}}) / m_{\text{best-causal}}$$

and, separately, $\text{union} / \text{logical}$.

Two conventions deserve emphasis. First, m uses the **pre-deduplication** denominator so that a single number remains comparable across concurrency levels; the deduplication benefit itself is reported as $\text{union} / \text{logical}$ and is never counted as a policy’s contribution, because any FCFS scheduler obtains it for free. Second, g is a *relative* quantity and is meaningful only alongside the absolute traffic; §6.1 shows that a small g can indicate either that existing policies suffice or that no policy can help, and that the two are distinguished only by absolute transferred bytes against a bandwidth budget.

For a static pinned set, the initial load of the pinned blocks is counted in M and is reported separately from steady-state traffic. Because this one-time cost is sensitive to trace length, static/dynamic comparisons must also report the fit source, quota rule, number of events, and either total traffic or an amortized steady-state view.

2.5 Reference trace

The artifact includes a deliberately small trace — 3 layers, 4 experts per layer, 5 scheduling steps, 30 union accesses — that can be replayed by hand. Under Definition 2 with the released capacity setting it yields 12 hits and 18 misses. It is intended as an executable specification: a simulator that does not reproduce these counts does not implement the semantics used in this paper.

§3 Experimental Setup

3.1 Why expert caching is a cache problem with unusual structure

A routed MoE layer selects k of N experts per token; under offloading the resident subset lives in device memory and the remainder is streamed on touch. Three properties shape every measurement that follows. The access unit is a set, not an element: all experts a step touches at a layer are required together (§2.1), so hit/miss classification is naturally event-level. Concurrency deduplicates for free: experts selected by several concurrent requests are fetched once, a benefit belonging to the scheduler rather than to any policy, which we account for separately throughout. And the binding budget is bandwidth, not capacity alone: raising the residency fraction lowers traffic but consumes memory that also serves KV state and workspace, so a policy is useful only if it reduces bytes at fixed residency, or permits lower residency at fixed service quality.

3.2 Models

	experts N	top- k	MoE layers L	blocks $L \cdot N$
Granite-3.1-3B-A800M	40	8	32	1,280
OLMoE-1B-7B-0125	64	8	16	1,024
Qwen3-30B-A3B (4-bit)	128	8	48	6,144

The model-family descriptions follow their official technical reports or model documentation ([IBM Research 2024](#); [Muennighoff et al. 2025](#); [Yang et al. 2025](#)).

The exact model revisions are frozen in the route manifests: Granite commit `d4dd87aa3a6c201bc374851d7d7ff4cf39a0b82a`, OLMoE commit `9b0c1aa87e34a20052389dce1f0cf01da783f654`, and the MLX Qwen3 snapshot `d388dead1515f5e085ef7a0431dd8fadf0886c57` (4-bit, group size 64).

Qwen3-30B-A3B is the primary model: its expert count places it at the boundary where the per-step expert union crosses per-layer cache capacity (§6), which is the regime the two smaller models cannot reach. The smaller models are used for cross-model checks and for the replay-semantics measurement of §4. All three are evaluated at the same residency fractions and under identical replay semantics.

Analytical reference frame. To express results in engineering units we also use a paper specification of a frontier-scale MoE: 92 MoE layers, 896 routed experts, top-16, 82,432 expert blocks of approximately 16.74 MiB, approximately 106.55 GiB of non-routed weights, approximately 1,347.12 GiB of routed expert weights, and approximately 25.83 GB of logical expert bytes per output token before deduplication. **No trace from this model was collected and no measurement in this paper was taken on it.** It is used only to state operating regimes and unit conversions, and every quantity derived from it is labelled as an analytical projection, never as a prediction.

3.3 Traces

Decode routing was collected autoregressively with the model’s chat template applied and reasoning mode disabled, greedy decoding, prompts capped at 4,096 tokens, and up to 384 recorded decode forwards per request with natural end-of-sequence termination. For each request we store the emitted token sequence, the effective length, and the complete per-layer router logits, top- k indices, probabilities, gate weights, entropy and margin.

Two collection choices matter for later sections and are the direct consequence of §5. Applying the chat template prevents the model from continuing the instruction text instead of answering it. Recording several hundred decode steps places the majority of each trace beyond the boilerplate-dominated opening region. An earlier collection that did neither is retained and re-analysed in §5 as the contaminated condition.

Traces are converted into event streams (§2.1) under first-come-first-served continuous batching with same-step cross-request deduplication. Admission offsets are configurable so that concurrent requests are not locked to identical decode positions.

The primary Qwen3 collection ran on a Mac mini with an Apple M4 Pro and 24 GiB unified memory under macOS 26.5.1, Python 3.12.13, MLX 0.32.0 and MLX-LM 0.31.3. It contains 144 confirmatory requests in 24 shards, uses the snapshot above, and reached 19.31 GB peak Metal allocation. All collection parameters and shard hashes are stored in the source manifest rather than inferred from filenames. MLX is cited as versioned software rather than as an inferred hardware-system paper (MLX Contributors 2025).

3.4 Workload probe set

Measuring workload-conditioned routing requires requests partitioned by task. We use **ControllerProbe-D1**: 432 records over six workload archetypes — document RAG, tool agent, ERP structured analytics, office/legal, process diagnostics, and equipment maintenance/BOM — built from public benchmark and public industrial sources. The benchmark-derived portions use LongBench, Berkeley Function Calling Leaderboard, Spider 2.0, and LegalBench (Bai et al. 2023; Yan et al. 2024; Lei et al. 2024; Guha et al. 2023); the industrial portions include the Petrobras 3W data and Apache OFBiz sample entities (Vargas et al. 2019; Apache Software Foundation 2026), with the complete source and licence ledger released alongside the artifact. Its construction is described in §5; the properties that matter here are:

- twelve structurally distinct prompt forms, each terminating in a different kind of line, crossed with two instruction languages, so that no two records in a cell share a form–language pair;
- eight task framings per archetype and five payload rendering styles;
- a **matched-pair control arm** in which a subset of the same source records is re-rendered with a single fixed template, so that template effects can be measured rather than argued;
- a group-aware discovery/confirmatory split with no source group in both sides.

Category proportions in the probe set are a design choice for controlled comparison and are **not** an estimate of any deployment’s traffic mix.

3.5 Simulator and statistical discipline

The simulator implements Definition 2 with per-layer and global scope, seeded tie-breaking, and the policy set of §2.3. It is validated against the reference trace of §2.5.

Every threshold presented as a confirmatory **decision criterion** was pre-registered and frozen before the corresponding measurement was read, together with the split it would be evaluated on. Exploratory diagnostics, failure attribution, and reviewer-driven audits were not pre-registered. We release the decision documents unchanged, including those in which a criterion failed and the associated development line was stopped. Where a design parameter was revised, the revision and its rationale are recorded with the statement that no result had yet been generated or read; §8 gives one such case. Discovery and confirmatory splits are disjoint at the source-group level, and confirmatory data is read once.

We report tie-seed sensitivity for the principal conditions. We do not report population confidence intervals for quantities estimated from a single trace, and we mark such quantities as implementation-stable rather than statistically bounded.

§4 Axis I — Replay Semantics

Claim. Under the fused-event traffic contract of §2, replaying an event as individual accesses permits eviction and artificial refetch of an expert that was resident at event start and is still required later in the same event. The resulting error is not uniform noise: it penalizes recency-based policies by roughly 27–29% while leaving frequency-based, offline, and static policies within 4%, and it inverts the policy ranking.

4.1 The mechanism

Under sequential replay, early misses may be admitted before all start-resident members of the event have been served. If an early admission evicts a resident expert that appears later in the flattened order, that later access is counted as a miss even though the fused-event contract would have counted it as a hit. Evicting an already-served expert does not create a current-event miss; it can, however, bias the event’s final retained set and therefore future events.

A recency-based policy is especially sensitive because the arbitrary order inside the event immediately changes its key and its victims. Frequency-based policies are less sensitive when their counters are dominated by prior events. The offline optimum uses future position, and a static pinned set never evicts. This difference in policy sensitivity, rather than a uniform counting offset, is the mechanism tested below.

Expert identifiers make this worse in a specific way. If the flattened order is the natural one — ascending expert index — then the surviving residents are systematically the highest-indexed experts of each event, and expert numbering, an arbitrary artifact of the checkpoint, enters the retention decision.

4.2 Randomization is not a fix

A natural repair is to randomize the intra-event order. Table 1 shows that this addresses only half of the problem.

Table 1. Miss ratio under three replay semantics. (*64-expert model, per-layer scope, $\rho = 40\%$, LRU; range over 5 tie seeds.*)

replay semantics	miss ratio
sequential, ascending expert index	99.73%
sequential, seeded-random order	87.15% – 87.19%
event-atomic (Definition 2)	56.92% – 57.11%

Randomization removes the dependence on expert numbering, which accounts for the drop from 99.73% to roughly 87%. It does not remove intra-event eviction of not-yet-consumed residents, which accounts for the remaining 30 percentage points. Only deferring the retention decision to the event boundary removes both.

4.3 The distortion is selective, and it inverts rankings

Table 2 holds the trace, capacity, concurrency, cache scope and residency fraction fixed and varies only the replay semantics. It is the data behind Figure 1.

Table 2. Effective miss fraction under sequential and event-atomic replay. (*128-expert model, held-out split, per-layer scope, $\rho = 40\%$, $B = 8$. Identical trace and capacity in both columns.*)

policy	sequential replay	event-atomic	inflation
LRU	24.59%	19.30%	+27.4%
LFRU	23.14%	18.01%	+28.5%
Least-Stale	24.59%	19.30%	+27.4%
LFU	19.80%	19.14%	+3.4%
Belady	10.36%	9.93%	+4.3%
Static-same-trace (diagnostic)	19.10%	19.10%	0.0%

Two observations follow.

The ranking inverts. Under sequential replay the best causal policy is the same-trace static diagnostic, which appears to beat LFRU by 4.04 percentage points. Under event-atomic replay LFRU is best and beats

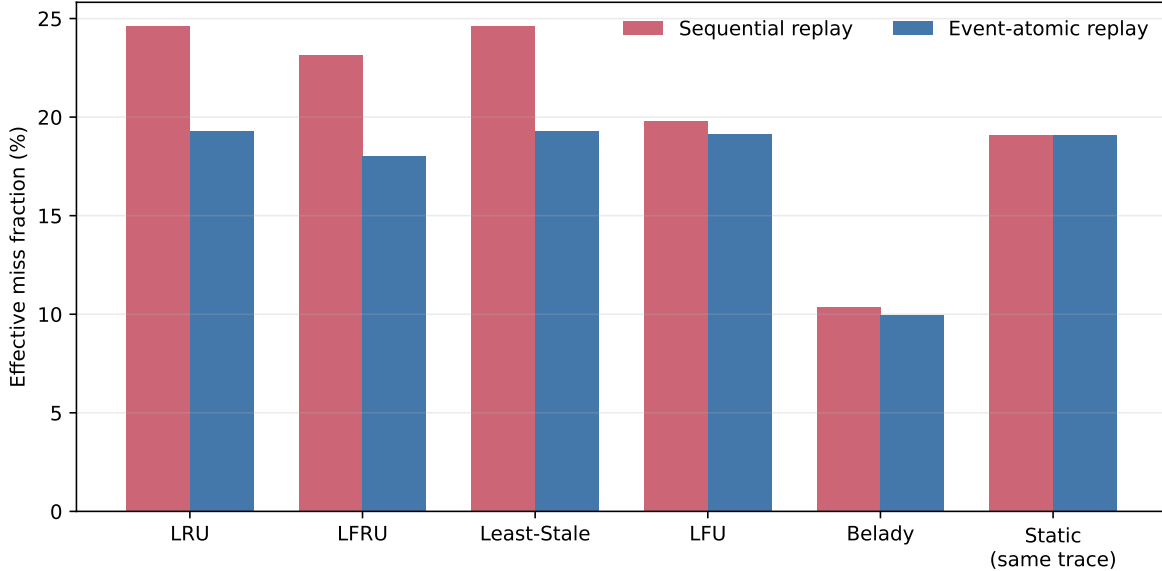


Figure 1: **Figure 1. Replay semantics selectively changes policy measurements.** The trace, capacity, concurrency, scope and residency fraction are identical; only replay semantics changes. Values are built from the frozen Table 2 artifact.

that diagnostic by 5.70% relative (1.09 percentage points). A study that concluded “a fixed, offline-chosen residency set outperforms online replacement” would be an artifact of the replay model alone.

The static row is fitted on the evaluated trace in both columns so that replay semantics is the only changed variable. It is therefore evidence of replay-invariance, not evidence for a deployable static policy; §9 evaluates the discovery-frozen protocol separately.

The error cannot be treated as a calibration offset. Because the inflation ranges from 0.0% to 28.5% across policies, no single correction factor recovers the correct ordering. Results obtained under sequential replay are not convertible to event-atomic results after the fact.

Least-Stale coincides exactly with LRU in both columns. This is expected rather than anomalous: in the absence of prefetching, the two-generation staleness test degenerates to last-access order. We retain it as a distinct implementation so that the diagnostic is explicit, and we note that its published advantage is reported in combination with prefetching and miss handling, which we do not evaluate here.

4.4 Consequence for a downstream conclusion

The semantics changed one of our own conclusions rather than merely its precision. Under event-atomic replay, the steady-state prefill gap between the best causal policy and the offline optimum is:

Table 3. Recoverable gap, prefill, per-layer scope, mean over 5 tie seeds.

model	$\rho = 20\%$	$\rho = 30\%$	$\rho = 40\%$
40-expert	0.155%	0.320%	0.733%
64-expert	0.601%	1.516%	3.169%

All six points fall far below the 10% threshold we had pre-registered as the level at which designing a new steady-state prefill eviction rule would be worthwhile. Under the earlier sequential semantics the same measurements had appeared to leave several times that much headroom, and a conclusion we had drawn

from them — that admission and pinning were the dominant mechanism — did not survive the correction and was formally withdrawn.

We report this because it illustrates the failure mode the paper is about. The sequential result was not obviously wrong when produced: the miss ratios were plausible, the policies were implemented correctly, and the relative ordering looked reasonable. The defect was in the replay model, which is exactly the component least often described.

The magnitude of the inflation is not a constant: it grows with concurrency and shrinks with residency fraction, converging when events fit inside the quota. The qualitative property we claim — selectivity by policy family, and therefore ranking change — follows from the contract argument of §2.2 and does not depend on the operating point (§11).

§5 Axis II — Workload Contamination

Claim. Probe sets that wrap each workload category in a single instruction template produce verbatim-identical generation prefixes across concurrent requests. The resulting expert overlap is readily mistaken for semantic locality. A matched-pair rendering intervention changes the measured early-window effect by 19.4–31.9 percentage points, and correcting the construction reverses the point-estimate category ordering.

5.1 A conclusion that did not survive

We first measured workload-conditioned expert locality on a probe set built the obvious way: six workload archetypes, one instruction template per archetype, public payloads substituted into each. The result was clean, passed a pre-registered confirmatory split, and had an appealing interpretation — industrial process-diagnostics traffic appeared to concentrate on a markedly narrower expert working set than general office traffic.

Table 4 shows what happened when the probe set was rebuilt to control for prompt-surface repetition, holding the model, the payload sources, the cache configuration, and the metric fixed.

Table 4. Reduction in effective miss fraction of a homogeneous single-archetype stream relative to a mixed stream. (*128-expert model, per-layer scope, $\rho = 40\%$, $B = 8$, event-atomic.*)

workload archetype	single-template probe set	diversity-controlled probe set
process diagnostics	57.2% (<i>headline result</i>)	5.9%
document RAG	57.9%	43.7%
ERP structured analytics	36.2%	7.6%
equipment maintenance / BOM	35.3%	22.0%
office / legal	26.9%	53.3%
tool agent	9.5%	10.2%

The point-estimate ordering does not merely compress; it inverts. The archetype that had supported the strongest claim falls to near the bottom, and the archetype that had ranked second from last rises to the top. We withdrew the original conclusion. The remainder of this section explains why it was wrong, how to detect the failure, and how much of the corrected ordering is actually supported.

5.2 Mechanism

Three conditions combine, none of which is individually unusual.

A shared template tail. Every prompt in a category ends with the same instruction sentence. Under raw continuation — that is, when the model’s chat template is not applied — the model completes that sentence verbatim before producing anything task-specific, and then emits a fixed reasoning preamble.

A short decode window. If only a few dozen decode steps are recorded, the shared prefix is a large fraction of every trace. In our original collection the first 15–25 of 63 recorded steps were near-identical across requests.

Position-locked cohorts. If all requests are the same length and are admitted and retired in lock-step, concurrent requests occupy identical decode positions, so the shared prefix aligns exactly at the same scheduling step. The per-step expert union is then computed over near-identical token contexts.

The consequence is direct. In one archetype, 4 of 16 requests produced byte-identical 63-token outputs; positional token agreement within that archetype was 20.65%, against 1.04% for a category whose payloads were long heterogeneous documents.

5.3 Two diagnostics that separate the explanations

Within-category expert overlap alone cannot distinguish semantic locality from surface repetition. Two measurements do, and neither requires a control arm.

Residual after removing near-duplicate text. We restrict the comparison to request pairs whose generated text barely overlaps — positional token agreement below 5% — and compare the residual against a cross-archetype baseline.

Table 5. Within-archetype pairwise expert Jaccard, all pairs and restricted to low-text-overlap pairs. (*16 requests per archetype.*)

single-template set			diversity-controlled set	
archetype	token agr.	expJac lowTok	token agr.	expJac lowTok
process	20.65%	0.158	0.78%	0.087
diagnostics				
equipment / BOM	15.33%	0.092	1.06%	0.108
ERP	5.15%	0.118	0.61%	0.096
tool agent	1.85%	0.103	0.53%	0.085
document	1.04%	0.144	0.45%	0.119
RAG				
office / legal	0.74%	0.123	0.57%	0.144
<i>cross-archetype baseline</i>		<i>0.078</i>		<i>0.077</i>

On the single-template set, equipment/BOM has a raw within-class Jaccard of 0.229; restricting to low-text-overlap pairs collapses it to 0.092, essentially the cross-archetype baseline of 0.078, and process diagnostics falls from 0.291 to 0.158. Categories whose payloads were naturally heterogeneous barely move. On the diversity-controlled set all sequences are unique and the two columns nearly coincide, indicating that the surface-repetition channel has been closed rather than reduced. We report **expJac | lowTok** minus the cross-archetype baseline as the residual workload effect.

Prefix-aligned decay. Template echo must fade as the shared prefix ends. A genuine task effect can also vary across response phases, so position dependence alone is neither necessary nor sufficient; the stronger signal is an abrupt decay aligned with a shared, near-identical prefix.

Table 6. Reduction in per-step expert union of a homogeneous stream relative to a category-round-robin mixture, by decode-position band. ($B = 16$.)

archetype	single-template set		diversity-controlled set	
	early (t 0–8)	late (t 32–63)	early (t 0–16)	late (t 160–384)
process diagnostics	65.7%	29.1%	6.0%	−7.8%
equipment / BOM	39.2%	13.8%	3.7%	6.2%
ERP	24.5%	11.5%	6.8%	−8.6%
document RAG	1.2%	20.2%	7.1%	17.1%
office / legal	4.0%	9.7%	11.6%	20.1%
tool agent	7.1%	5.8%	−8.6%	24.3%

On the contaminated set the three template-rendered categories decay steeply while the two with heterogeneous natural-language payloads are flat or rising; on the corrected set the decaying categories collapse to near zero or below and the rising ones persist. Combined with prompt and output overlap this gives a screening rule: **an effect concentrated in a position band occupied by a shared generated prefix should be treated as surface repetition until a diverse or matched control says otherwise.**

5.5 Matched-pair attribution

Diagnostics indicate contamination; they do not quantify it. We therefore render a subset of the *same source records* twice — once with the diversified forms and once with a single fixed template ending in a fixed instruction — and pair them by source record. Model, cache configuration, concurrency and metric are identical; the prompt-rendering policy is the treatment.

Table 7. Difference in apparent union reduction under the fixed rendering, matched on source record, early decode band (t 0–16).

archetype	template-attributable
office / legal	+31.9 pp
process diagnostics	+28.8 pp
ERP	+26.4 pp
tool agent	+24.5 pp
equipment / BOM	+20.9 pp
document RAG	+19.4 pp

This matched contrast is causal for the **rendering policy**, not for a pure full-service template-echo component: changing the wrapper also changes the continuation, response length and cache churn. Its early position-aligned band, together with the shared prefix, is what specifically supports the echo interpretation.

One asymmetry deserves note. The fixed-template arm has a *higher* effective miss fraction over the full trace (21.99% vs 18.78% at $\rho = 40\%$, $B = 8$) even though it shows a larger early-window union reduction, because the fixed template also shortens responses and increases cold-start frequency. Template echo therefore cannot be read off the end-to-end miss difference; only the position-resolved union comparison isolates it.

5.6 A diversity-controlled probe set

The corrected set holds payload sources fixed and varies only the wrapper: twelve structurally distinct prompt forms, each terminating in a different kind of line, crossed with two instruction languages, so that no two records in an (archetype, split) cell share a form-language pair; eight task framings per archetype; five payload rendering styles; and, for the industrial archetypes, six source families with different record shapes. Candidate payloads whose 5-gram Jaccard against an already-selected payload exceeds 0.30 are rejected at selection time, because several public sources are internally repetitive. The matched-pair arm of §5.5 is built in. The result is 23-24 distinct 70-character prompt heads per 24-record cell against 1-4 per 20-record cell in the original, and a mean pairwise 5-gram Jaccard of 0.002-0.014 against 0.038-0.190.

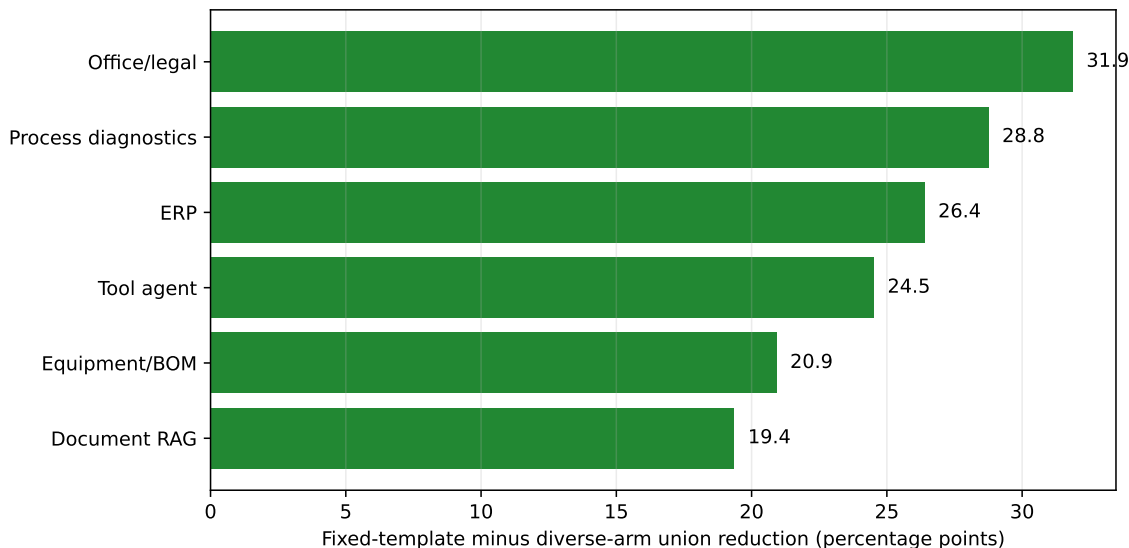


Figure 2: **Figure 2. Prompt rendering contributes materially to the apparent early-window workload effect.** Each bar is the fixed-template arm minus the diverse arm for the same source records at $t=0-16$; it is not an end-to-end miss reduction.

5.7 How much of the corrected ordering is supported

Correcting a measurement invites the assumption that the corrected result is sound. We checked. Two issues affect Table 4’s right-hand column.

First, the mixed reference stream contains six times as many distinct requests as each homogeneous stream, and a stream with fewer distinct requests has more inter-request reuse for reasons unrelated to workload. We rebuilt the reference as seven size-matched draws. The confound proves small — the size-matched reference is 17.81% against 18.76% for the original — but the seven draws span 15.54%–19.04%, a standard deviation of 1.18 percentage points.

Second, each homogeneous condition is itself a single draw of twelve requests. We replicated all six on a disjoint held-out set of twelve.

Table 8. Homogeneous-stream miss fraction, two independent draws, against a size-matched reference. Interval propagates the reference’s draw-to-draw spread; it is not a population confidence interval.

archetype	draw 1	draw 2	mean	improvement vs 17.81%	± 1 s.d. of reference
office / legal	8.76%	8.15%	8.45%	52.5%	49.2 – 55.5
document RAG	10.57%	10.69%	10.63%	40.3%	36.1 – 44.0
equipment / BOM	14.64%	13.33%	13.99%	21.5%	15.9 – 26.4
ERP	17.33%	15.18%	16.25%	8.8%	2.3 – 14.4
process diagnostics	17.65%	16.34%	16.99%	4.6%	–2.2 – 10.5
tool agent	16.85%	18.17%	17.51%	1.7%	–5.3 – 7.8

Only the top two archetypes are clearly separated from the reference; equipment/BOM is positive with a wide interval; and the bottom three are not distinguishable from zero at this sample size. The honest

statement is therefore not that six archetypes form an ordering, but that **two of six show a workload effect that survives both the contamination correction and a size-matched reference.**

5.8 Applicability beyond our own data

The diagnostics of §5.3–§5.4 are not specific to our probe set. Applied to the first 48 complete records of a public Mixtral routing artifact over C4 input sequences, each truncated to 512 input tokens, they report 48 of 48 unique token sequences and a 95th-percentile positional agreement of 1.37% — no contamination signal (Raffel et al. 2020; Allen Institute for AI 2024). As a labelled sensitivity check, we then copied one record’s first 64 token-and-route positions into every record. Positional agreement rose from 0.641% to 13.050%, mean expert Jaccard from 0.185 to 0.286, and the diagnostic flag changed from false to true.

The natural-C4 run is a false-positive check and the injected run is a synthetic positive control. Neither is an external replication of the prompt-rendering effect: the public artifact contains no workload labels or matched template arm, and the injection is not a model-generated response.

The effect requires a shared template tail, a decode window short enough for the prefix to dominate, and, for the union metric, position-aligned concurrency; we do not claim every template-based probe set satisfies them (§11).

§6 Axis III — Operating Regimes

Claim. Normalized miss fractions do not transfer across MoE models. The distribution of the per-step, per-layer expert union relative to cache capacity is a first-order variable that must be reported before any cross-model comparison is meaningful. Its mean is necessary but not sufficient: holding the event-set multiset and mean ratio exactly constant and permuting only temporal order moves the offline-optimal gap from 44.9% to 30.8% and changes which causal policy is best.

6.1 The regime variable

Two quantities determine whether a replacement policy has room to act at all: how many distinct expert blocks a single event requires, and how many the layer can hold. Write

$$r(s, l) = |E(s, l)| / c_l, \text{ and } r_bar \text{ is its mean over reported events.}$$

For an event with $r(s, l) < 1$, its working set fits in the quota; misses can still arise from drift between events and eviction order may matter. For an event with $r(s, l) > 1$, at least $|E| - c_l$ members cannot remain resident after the event, creating a capacity-forced floor. As the mass of the $r(s, l)$ distribution above one grows, replacement policies have less headroom. This is a tendency, not a theorem that all policies or all traces converge.

r_bar is not a free parameter. It generally rises with concurrency and top-k, and falls with expert count and residency fraction, so two models compared “at the same batch size” are routinely in different regimes.

6.2 Crossing the boundary

Table 9 sweeps concurrency on the 128-expert model at fixed residency, reporting the measured union, the fraction of events that exceed the layer quota, and the resulting gap.

Table 9. Regime and gap versus concurrency. (*128-expert model, per-layer scope, $\rho = 40\%$, c_l approximately 51.2, event-atomic.*)

B	union / layer-step	p95	r_bar	events over quota	recoverable gap
1	8.00	8	0.156	0.00%	46.97%

B	union / layer-step	p95	$\mathbf{r_bar}$	events over quota	recoverable gap
2	15.02	16	0.293	0.00%	49.25%
4	26.58	30	0.519	0.00%	48.12%
8	42.63	50	0.833	2.19%	44.96%
16	59.21	72	1.156	81.49%	33.32%

The gap is roughly flat while $\mathbf{r_bar} < 1$ and falls sharply once the union routinely exceeds capacity. The measured unions also fall below an independent uniform routing null by 3.1% at $B = 2$ rising to 28.2% at $B = 16$, so concurrent requests do overlap; the crossover is nevertheless reached between $B = 8$ and $B = 16$.

6.3 Same model, same workload, capacity only

Concurrency is not the only way to move $\mathbf{r_bar}$. Holding the model, the trace and the concurrency fixed and varying only the residency fraction produces the same effect.

Table 10. Gap versus residency fraction at fixed concurrency. (*128-expert model, per-layer scope, $B = 16$.*)

ρ	best causal	Belady	recoverable gap
20%	30.33%	28.08%	7.42%
30%	22.24%	18.09%	18.65%
40%	15.58%	10.39%	33.32%

A study reporting only the $\rho = 20\%$ row would conclude that existing policies are within 7.4% of optimal and that the problem is closed. A study reporting only the $\rho = 40\%$ row would conclude that a third of the traffic is recoverable. Both are the same model, the same workload and the same concurrency.

6.4 Cross-model comparison fails on batch size and improves on $\mathbf{r_bar}$

Table 11 compares three models two ways: aligned on batch size, as is conventional, and aligned on mean $\mathbf{r_bar}$.

Table 11. Recoverable gap under two alignment choices. ($\rho = 40\%$, *per-layer scope, event-atomic.*)

alignment	model	B	$\mathbf{r_bar}$	gap	spread
on batch size	40-expert	8	1.75	8.17%	36.8 pp
	64-expert	8	1.44	28.26%	
	128-expert	8	0.83	44.96%	
on $\mathbf{r_bar} \approx 0.30$	64-expert	1	0.313	48.13%	1.1 pp
	128-expert	2	0.293	49.25%	
on $\mathbf{r_bar} \approx 0.51$	40-expert	1	0.500	43.48%	4.6 pp
	128-expert	4	0.519	48.12%	

Aligned on batch size, the three models appear to behave qualitatively differently — one suggests the problem is essentially closed, another that it is wide open. Aligned on $\mathbf{r_bar}$, the same measurements agree to within a few percentage points. The apparent architecture dependence was a regime difference.

This also bounds what small models can be used for. The 40-expert model at $\rho = 40\%$ cannot reach $\mathbf{r_bar} < 0.5$ at any concurrency, because a single request already touches 8 of its 16 per-layer slots. Regimes below that are simply not observable on it, and results from it cannot be extrapolated into them.

A note on the analytical reference frame. For the frontier-scale specification of §3.2 — 896 experts, top-16, $\rho = 40\%$, giving $c_1 \approx 358$ — the expected per-step union at $B = 8$ under a uniform routing null is approximately 120, so $r_{\text{bar}} \approx 0.335$ under that null. That places it near the 64-expert model at $B = 1$ and the 128-expert model at $B = 2$, and far from any of them at $B = 8$. We state this only to illustrate that the alignment matters at scale; it is an analytical projection from published architecture parameters, not a measurement.

6.5 Necessary but not sufficient

r_{bar} is a normalizer, not a sufficient statistic. To show this directly we hold the multiset of event expert sets and r_{bar} **exactly** constant — the same events, the same sets, the same mean value 0.8237 — and permute only their temporal order.

The recoverable gap moves from 44.85% to 30.79%–31.06%, and the best causal policy changes from LFRU to LFU.

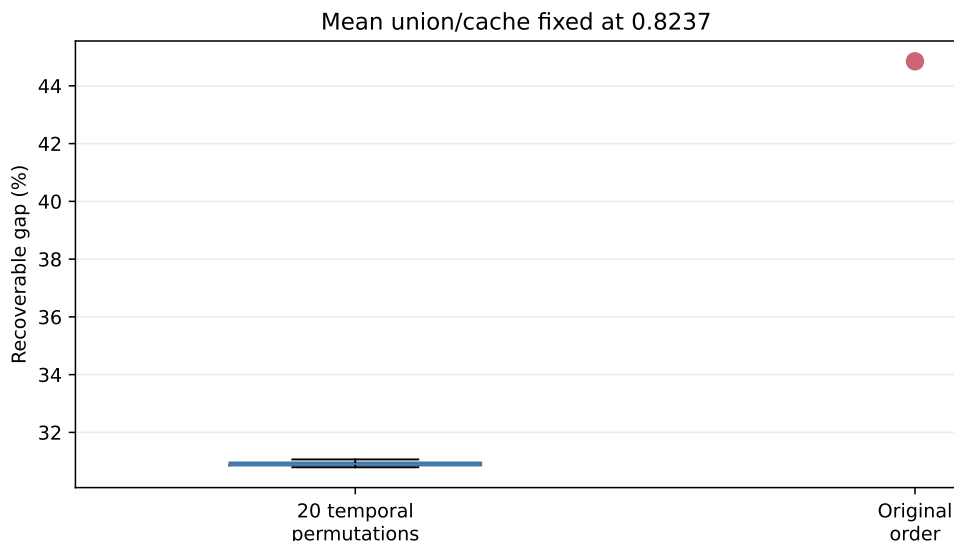


Figure 3: **Figure 3. Matching mean union/cache is not sufficient.** The original event order and all twenty permutations have the same event-set multiset and mean ratio (0.8237); only temporal order changes.

The permutation does not represent a realizable request scheduler; it is a mathematical sufficiency counterexample. It shows that reuse structure, not just working-set size relative to capacity, determines both the size of the gap and which policy realizes it. Quantities that vary independently of r_{bar} and that we expect to matter include the inter-event drift of the working set, the reuse-distance distribution, expert popularity concentration, top- k/N , cache scope, and the scheduling discipline. §9 gives an instance of the last: changing only the service discipline, at fixed request set, capacity and mean batch size, moves a static pinned set from 7.5% worse than LFRU to 17.0% better.

The correct prescription is therefore: **report the distribution of $r(s,1)$ and at least r_{bar} — without it, cross-model comparison is not interpretable — but do not treat matching r_{bar} as sufficient grounds for extrapolation.**

6.6 Consequence: gap alone is not a decision criterion

Sections 6.2 and 6.3 have a practical corollary. A small gap is ambiguous: it can mean that existing policies are close to optimal, or that the system is in the thrashing regime where nothing helps. The two are distinguished only by the absolute traffic relative to a transfer budget.

Here “offline traffic” means Belady’s transferred bytes under the fixed access sequence, because causal traffic above budget does not by itself establish physical infeasibility.

	small causal-to-offline gap	large causal-to-offline gap
offline traffic within budget	existing policies suffice; a controller is unnecessary	genuine algorithmic headroom on the fixed sequence
offline traffic over budget	no residency policy on this sequence is viable	oracle also fails; the gap is not enough to cross the budget

We recommend that any reported gap be accompanied by the absolute transferred bytes per output token and the bandwidth budget implied by the target service rate. In our own use of this criterion, the smaller models produced operating points in the lower-left cell — gap under 10% with the offline optimum still several times over budget — that a gap-only reading would have classified as “existing policies already suffice.”

We do not claim $\mathbf{r} = 1$ is a sharp threshold, only that behaviour changes across it; the permutation ablation establishes insufficiency rather than quantifying how much of the gap reuse structure explains in general (§11).

§7 The Offline-Optimal Gap and Its Decomposition

Claim. With all three axes controlled, a large gap to the offline optimum remains and is stable across workload composition. Decomposing it shows that 84.3–96.6% derives from knowing which resident block is used furthest in the future, a quantity for which the natural online estimator not only fails to help but performs worse than the baseline it replaces. A large offline-optimal gap is therefore not, on its own, evidence of an online opportunity.

7.1 The gap is large and stable

Table 12. Recoverable gap on the principal conditions. (*128-expert model, per-layer scope, $\rho = 40\%$, event-atomic; best causal chosen from LRU, LFU, LFRU, Least-Stale.*)

condition	best causal	Belady	gap
held-out, $B = 8$	18.01% (LFRU)	9.93%	44.85%
held-out, $B = 2$	12.84% (LFRU)	6.37%	50.42%
calibration, $B = 8$	18.76% (LFRU)	10.35%	44.90%
fixed-template arm, $B = 8$	21.99% (LFRU)	12.02%	45.33%

The gap does not depend on the contamination of §5: the fixed-template arm, whose absolute miss fraction is 22% higher, shows the same gap to within half a percentage point. Nor is it an artifact of a particular workload mixture. Holding $\rho = 40\%$, $B = 8$ and the replay semantics fixed, we constructed thirteen streams — seven size-matched mixtures drawn from disjoint request pools and six homogeneous single-archetype streams — and measured the gap on each.

Table 13. Gap invariance across workload composition. (*13 deterministically frozen conditions, $\rho = 40\%$, $B = 8$, per-layer scope. The conditions comprise six matched mixtures, one non-matched mixture, and six non-matched pure streams.*)

	gap
minimum	44.18%
maximum	45.93%
mean	45.09%

The best causal policy is LFRU in twelve of thirteen conditions and LFU in the homogeneous office/legal stream. **This range applies to $\rho = 40\%$, $B = 8$ only.** It must not be pooled with the $B = 2$ condition of Table 12 or with the residency sweep of §6.3, whose gaps span 1.95%–44.78%; those are different operating points and combining them would misrepresent the variability.

Across the four principal conditions, varying the tie-breaking seed moves the gap by order 10^{-5} . We report this as implementation stability, not as a population confidence interval: it bounds the simulator’s nondeterminism, not the sampling variability of the underlying traffic.

7.2 What the offline optimum’s advantage consists of

Belady’s advantage over a causal policy has two distinguishable sources.

Bypass admission. On a miss, the optimum may decline to cache the incoming block at all, if that block’s next use is farther away than that of every resident candidate. This has familiar causal approximations in storage caching, including admission filters and reuse prediction, but the oracle share measured below is not thereby guaranteed to be recoverable.

Future-victim selection. When the optimum does admit, it evicts the resident block whose next use is farthest away. This asks for a *ranking* of all resident blocks by next-use distance, a substantially harder estimation problem.

Both use future information; they differ in how tractable their online analogues are. To separate them we run a third variant, **Belady-forced-admit**, which retains furthest-next-use eviction but is required to admit every miss. It is not a deployable policy — it still consults the future — but it isolates the contribution of the admission decision. Writing m_c , m_B , m_F for the miss fractions of the best causal policy, Belady, and forced-admit Belady:

$$\begin{aligned} \text{admission share} &= (m_F - m_B) / (m_c - m_B) & \text{future-victim share} &= (m_c - m_F) / (m_c - m_B) \end{aligned}$$

Table 14. Decomposition of the recoverable gap. (*128-expert model, per-layer scope, $\rho = 40\%$, held-out split; discovery values in parentheses.*)

condition	total gap	admission share	future-victim share
$B = 8$	44.85%	15.69% (16.33%)	84.31%
$B = 2$	50.42%	3.40% (3.37%)	96.60%

The component with familiar online analogues is the minority in both conditions, and it shrinks as concurrency falls. At $B = 2$ — the operating point whose mean regime ratio $\mathbf{r_bar}$ is closest to the frontier-scale reference frame of §6.4 — 96.6% of the gap rests on ranking resident blocks by future distance.

7.3 Can the dominant component be estimated online?

The decomposition locates the gap; it does not establish that the located quantity is unpredictable. We tested that directly by replacing the oracle’s input with a prediction and leaving everything else unchanged.

We train a next-use-distance predictor on features available at the moment of each access: log recency, a first-observation indicator, log frequency, the log of the previous inter-access gap, the gate mass of the access, the log of the number of concurrent requests routing to that block in the same event, the layer index, and a popularity rate. The last three are not used by any of our causal baselines, so the predictor has strictly more information than LFRU. It is fitted on one trace and evaluated on a disjoint one, and its output is substituted

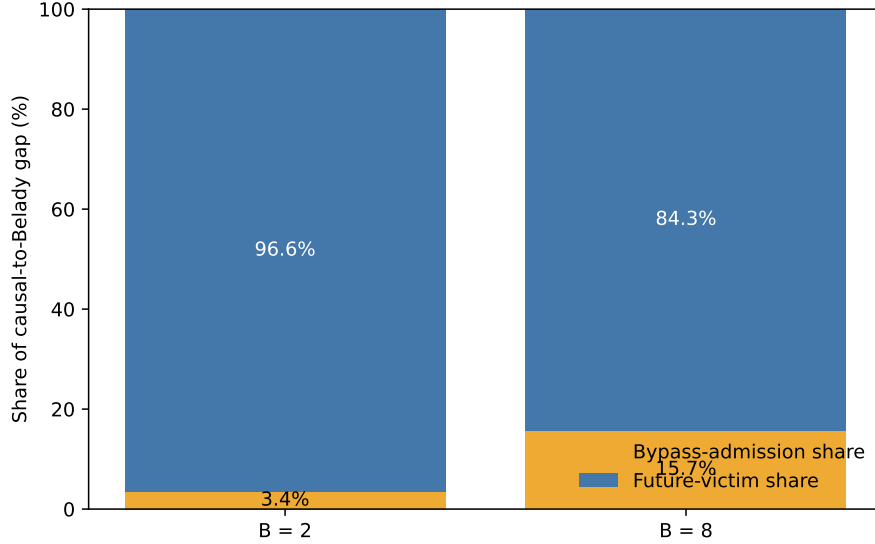


Figure 4: **Figure 4. Most of the offline-optimal gap is future-victim knowledge in the two held-out operating points.** Shares use event-atomic replay and the forced-admission decomposition; neither component is thereby guaranteed to be causally recoverable.

for the true next-use array in the *same* eviction and admission machinery, yielding an algorithmically causal candidate. We do not measure its inference cost or establish that it meets a production scheduling budget.

Table 15. Predicted-next-use policy against the baselines it replaces. (*128-expert model, fitted on calibration split, evaluated on held-out split, per-layer scope, $\rho = 40\%$, $B = 8$.*)

	effective miss fraction
LFU	19.14%
LFRU (best causal)	18.01%
predicted next-use	18.93%
Belady	9.93%
fraction of the gap recovered	−11.39%

The predictor is informative as a regressor — its transfer R^2 on log next-use distance is 0.237 — and still makes the policy worse than the baseline it was intended to improve. Regression accuracy over all accesses is, however, the wrong measure: eviction needs a correct *ordering* among the blocks that are resident at the moment of decision. We therefore instrumented the policy’s eviction decisions directly.

Table 16. Victim-ranking quality at eviction decision points, tie-aware. (*Same configuration; 1,708,374 eviction decisions, 46,172 sampled every 37th decision; mean candidate set 51.2 blocks; mean optimal-victim set 1.24 blocks; 95% CIs by cluster bootstrap over 3,055 scheduler steps.*)

victim chosen by	in the optimal-victim set	distance regret
random candidate	2.42% [2.28, 2.56]	0.00530
predicted next use	3.39% [3.22, 3.57]	0.00519
LRU	20.64% [20.20, 21.07]	0.00315
LFRU	22.11% [21.68, 22.56]	0.00301

rank correlation, predicted vs. true next use	Spearman
all resident candidates (average ranks for ties)	−0.207 [−0.210, −0.204]
excluding candidates never used again	−0.206 [−0.209, −0.203]

Because the optimal victim need not be unique, we score a choice as correct if it attains the maximum true next-use key, not if it equals a particular tie-broken identifier; Spearman uses average ranks. Ties are in fact rare here — the mean optimal set holds 1.24 blocks and on average 0.07 candidates are never used again — so neither convention drives the result.

Three readings agree. The predicted ordering is *anti-correlated* with the truth, and this survives both tie-aware ranking and dropping never-again candidates. The predictor selects an optimal victim 3.39% of the time, barely above the 2.42% obtained by choosing a resident block uniformly at random, and roughly six times less often than LRU or LFRU at the identical cache state. Its distance regret is statistically close to random and about 70% worse than either causal baseline. The predictor therefore supplies no usable ordering signal where the decision is taken, which is what the end-to-end result of Table 15 reflects; because the same predictions also drive bypass admission, errors compound, as a block whose next use is over-estimated is refused admission and then immediately required.

A positive global R^2 and a negative conditional rank correlation can coexist. Selection induced by the policy is one possible explanation — the resident set is not a random sample of blocks, since survivors are those the predictor scored as soon to be reused — but we have not isolated this experimentally, and we offer it as a hypothesis rather than a finding.

7.4 What this establishes, and what it does not

We separate three claims and do not permit inference between them.

	claim	status
1	The full gap is the offline bound for residency policies on a fixed access sequence	definitional
2	Bypass admission contributes 3.4–15.7%; future-victim selection contributes 84.3–96.6%	measured, stable across splits
3	The causal approximation of §7.3 fails to recover the gap	specific to these features, this predictor, this model, this workload, this operating point

Claim 3 does not imply that future-victim distance is unpredictable in principle, and we make no such claim. A predictor with materially higher transfer accuracy, or one using features our traces do not contain, could change the result.

It is also worth separating our negative result from a body of work that is sometimes read as adjacent. Expert *prefetching* predicts **which experts a future step will route to**; our predictor estimates **how long a cached block will remain unused**. These are different targets with different horizons and different error costs, and a positive result on the former is not evidence against the negative result here (Fang et al. 2025; Du et al. 2024; Gavhane et al. 2025; Zhu et al. 2026).

What we do claim is narrower and, we believe, portable:

In our evaluated settings a large offline-optimal gap substantially overstates the gains recovered by representative lightweight causal mechanisms.

The practical implication is a reporting requirement rather than a design conclusion. A gap reported without decomposition invites the reader to treat it as available headroom. We recommend that any work motivating a cache controller by an offline-optimal gap also report the forced-admission decomposition, so that the share of the gap resting on future-victim knowledge is visible.

The gap itself is regime-dependent (§6), so these values are not properties of MoE routing in general; the predictor is one model class and no architecture search was performed (§11).

§8 Scoped Negative Results and One Invalidated Design

Framing. §7 shows that the offline-optimal gap is dominated by a component whose natural online estimator does not help. This section reports what happened when we pursued four mechanisms that do not require estimating that component. Three produced scoped negative results; the hard-partition study had no power to test its intended mechanism because its dedicated cache was smaller than one request’s top-k set. We report each with the conditions under which it was measured, because at least one fails for a reason that does not generalize, and we do not claim mechanism-level closure.

8.1 The four attempted mechanisms

Each decision threshold below was frozen, together with the split it would be evaluated on, before the corresponding measurement was read. The later failure attribution and the determination that the partition grid was arithmetically incapable of a fair positive test are post-hoc audits, not confirmatory results.

Table 17. Pre-registered criteria and outcomes, with measurement scope.

mechanism	frozen measurement setting	pre-registered criterion	measured outcome in scope
steady-state prefill eviction / admission	40- and 64-expert models; $\rho = 20/30/40\%$; 128-token prefill blocks; strictly lossless	gap $\geq 10\%$	0.155–3.169%; not met
warm-cache workload transition	40- and 64-expert models; $\rho = 20/30/40\%$; directed $A \rightarrow B$ transitions; windows $N = 10/25/50/100$	gap $\geq 10\%$	1439/1440 units below 10%; not met
semantic category partitioning	128-expert model; $\rho = 40\%$; $B = 8$; FCFS; equal total capacity; 6 slots < top-k = 8	transfer reduction $\geq 5\%$	transfer increased 115.9%; invalid test of the mechanism because the design floor dominated
causal affinity batching	128-expert model; $\rho = 40\%$; $B = 8$ with 24 admitted; deadline rotation; $W = 4$; strictly lossless	transfer reduction $\geq 10\%$	4.76% (LFRU) / 4.55% (static); not met

The warm-transition experiment covers 1440 units — two models $\times$ three residency fractions $\times$ three request-order seeds $\times$ twenty directed transitions between five workload proxies $\times$ four post-transition windows. A single unit exceeded the threshold, at 10.74%, and decayed to 6.83%, 5.03% and 3.95% as the window widened from 10 to 25, 50 and 100 requests, which is the signature of a short-lived warm-start transient rather than a sustainable control opportunity.

8.2 One invalidated design

The partitioning experiment gave a dedicated per-layer slot budget to each of the two archetypes §5.7 identifies as genuinely cache-friendly, with the remaining four sharing a third partition, at exactly equal total capacity. Every allocation in a seven-point grid lost, and the frozen one increased transfer by 115.9%.

The natural reading — that partitioning destroys cross-category expert sharing — is largely wrong. Running the *shared* cache at each partition’s own slot count and weighting by each partition’s share of logical assignments predicts 37.30% against a measured 37.99%: **capacity fragmentation alone accounts for 96% of the degradation**. The cause is arithmetic. Top-k is 8, so a single request touches eight experts per layer, while the best dedicated partition holds six; no point in the grid could retain even one request’s per-step working set. **The experiment as designed could not have produced a positive result**, and we present it as an invalidated design and a measurement lesson (§10, item D5) rather than as a negative result for semantic partitioning. The arithmetic does not generalize: at the reference scale of §3.2, 358 per-layer slots divided six ways leave a 2.3–3.7x margin over the per-category per-step union, so the −115.9% figure must not be carried across. Appendix B gives the grid.

8.3 Affinity batching: not a prediction problem

The remaining mechanism does not touch residency at all. It changes which requests are co-scheduled, and therefore changes the event sequence itself — a degree of freedom that the Belady bound of §7 does not cover, since that bound is defined on a fixed access sequence.

Up to 24 sessions are admitted and at most $B = 8$ are served per step, with a frozen fairness constraint that an admitted request’s inter-service interval never exceeds $W = 4$ steps. Three schedulers share identical arrival, admission and completion rules and differ only in how they select up to eight requests from the active set: `fcfs_deadline`; `causal_prev_route`, which uses only each request’s per-layer expert set at its previous executed token and greedily minimizes the predicted union; and `oracle_current_route`, which runs the same selection using the true expert sets of the tokens about to execute. The oracle is a rolling, route-aware bound, not a global optimum, and all metrics are computed on the complete cache trajectory rather than per step.

Table 18. Causal affinity batching against FCFS, held-out split. (*128-expert model, $\rho = 40\%$, per-layer scope, $B = 8$, 24 admitted, $W = 4$.*)

policy	FCFS	causal	oracle	causal gain	oracle gain	share	$\Delta p99$
LFRU	23.08%	21.98%	21.71%	4.76%	5.92%	80.43%	+7.06%
static	19.15%	18.28%	17.64%	4.55%	7.90%	57.65%	+10.08%
pinned							

No request starved and the maximum inter-service interval was 4 throughout. The static-residency arm additionally violated the frozen p99 constraint at 10.08%.

The important number is the second-to-last column. The causal scheduler captures 80.4% of what the route-aware oracle achieves under the same fairness constraint. **The shortfall is therefore not a weakness of the prediction signal**; it is that the schedulable headroom under $W = 4$ at this concurrency is itself only 5.9–7.9%. Improving the predictor cannot reach a 10% threshold that the oracle does not reach either. This distinguishes the result from §7.3, where the causal estimator was the binding constraint.

The signal itself is real: within a request, the per-layer expert set at step $t+1$ overlaps that at step t with mean Jaccard 0.293, against 0.032 for two independent uniform draws — a ninefold enrichment. It is simply not worth much once fairness is enforced.

8.4 A methodological consequence: single-step order statistics

Before running §8.3 we estimated the headroom the cheap way, by sampling 300 random batch compositions at a fixed step and comparing the best to the mean union. That estimate was **19.3%**. The full-trajectory measurement under the same concurrency with $W = 4$ yields an oracle headroom of **5.9–7.9%**.

Two effects account for the difference and both are general. The maximum over 300 draws is an order statistic and is biased upward as an estimate of what a systematic method achieves. More importantly, a per-step measurement ignores conservation: requests that are deferred because they route poorly with the current batch must still be served later, and their cost reappears. A per-step union reduction is therefore an upper bound that can be almost entirely recovered by the later schedule.

We recommend that scheduling headroom for MoE caching be estimated on complete trajectories under the intended fairness constraint, and that per-step best-of- N figures not be reported as achievable gains.

8.5 What these results do not close

We state the complement explicitly, because the value of a negative result depends on its boundary.

- **Prefill execution ordering.** §8.1 finds little eviction/admission headroom in the measured prefill conditions only. Layer-major versus chunk-major prefill scheduling was never measured; under an independent uniform-routing approximation the two differ by roughly two orders of magnitude in total expert traffic for long contexts, and it remains the largest unexamined lever we are aware of.
- **Lossy settings.** All results assume strictly lossless service. Expert substitution, pruning, and reduced-precision fallback are excluded by construction.
- **Prefetching.** We evaluate residency and scheduling, not prefetch. §7.4 states why our negative result does not bear on expert prediction for prefetching.
- **Frontier scale.** Every mechanism was measured at r_bar values reachable by models of 40–128 experts. §6.4 shows that the reference frame of §3.2 sits at a mean regime those models reach only at low concurrency.
- **Other fairness settings.** §8.3 fixes $W = 4$ and 24 admitted sessions; relaxing either enlarges the schedulable headroom and was not swept.

8.6 Recorded design revision

One parameter in §8.3 was revised before execution. The initial draft admitted $B \times W = 32$ sessions; a pre-execution check showed this implies exactly 100% service load, leaving no discretion for affinity selection to exercise. The count was reduced to 24, and the revision was recorded together with the statement that no result had been generated or read, and with an explicit prohibition on relaxing W or reverting the count afterwards. The document is released unchanged.

§9 What Survives

Three corrections, three scoped negative results, and one invalidated design leave a small set of descriptive statements and baselines statements. This section reports them, together with one confounder we did not anticipate and found only because a mechanism experiment happened to use a different service discipline.

9.1 A static pinned set is a simple reference, not a frontier

The simplest possible residency policy holds a fixed set of experts chosen offline from a calibration trace and never updates it. It requires no counters, no eviction logic, and no runtime state. Table 19 compares it against dynamic policies under standard continuous batching, using a pin list fitted on the calibration split and frozen before the held-out split was read.

Table 19. Static pinned residency versus dynamic policies across concurrency. (*128-expert model, held-out split, per-layer scope, $\rho = 40\%$, event-atomic; pin list frozen from the calibration split; initial load counted.*)

active B	static pinned	LFRU	Belady	static vs LFRU
2	21.50%	12.84%	6.37%	−67.4%
4	20.81%	16.26%	8.43%	−28.0%
8	19.37%	18.01%	9.93%	−7.5%
16	17.10%	16.64%	11.32%	−2.8%
24	15.28%	14.92%	12.21%	−2.4%
32	13.84%	13.52%	12.20%	−2.3%

The static set never wins, but the margin narrows monotonically from a factor of 1.7 at $B = 2$ to 2.3% at $B = 32$. The mechanism is the regime variable of §6: at low concurrency the per-step working set is small relative to capacity and a dynamic policy can track it precisely, whereas a fixed set must cover the global popularity distribution; as concurrency grows the union approaches and then exceeds capacity and the advantage of tracking disappears.

At $B=8$ and above the pin list transfers with a modest absolute penalty, but at $B=2$ it is 67.4% worse than LFRU. This matters because a frontier-scale model at the same residency fraction may occupy a similarly slack union/capacity regime; our local data therefore do not support a product claim for static pinning.

The $B=8$ protocol audit makes the fit-source comparison explicit. On the same held-out event stream and with identical equal per-layer quotas, **Static-frozen** transfers 19.3684% of logical assignments, while the leaky **Static-same-trace** diagnostic transfers 19.1004%. The frozen list therefore pays 0.268 percentage points, or 1.403% relative, on this one held-out stream. Both numbers count the initial 2,458-block preload. We make no claim about an optimized nonuniform layer allocation: an earlier draft quoted such a result without a frozen artifact, so it is removed rather than reconstructed after the fact.

9.2 Static residency performance is a function of the popularity distribution alone

Because a static set never evicts, its steady-state miss count is simply the number of accesses to non-resident blocks. Measured over the held-out trace, the access distribution is only moderately concentrated:

fraction of blocks	share of accesses
top 10%	24.8%
top 20%	44.0%
top 30%	59.6%
top 40%	72.3%

For the **Static-same-trace** diagnostic at $\rho = 40\%$, the identity predicts a miss ratio of $1 - 0.723 = 27.99\%$ of **union accesses**, and the simulator reports 27.99% under that denominator before adding the one-time preload. This is not the pre-dedup effective-miss fraction used in Table 19. The identity is exact for the union-access denominator: static residency traffic depends only on the access-count distribution and not on access order.

This has a useful consequence. Unlike every other policy in this paper, the performance of a static pinned set can be computed from an **aggregate per-layer, per-expert access histogram**, with no sequential trace required. Predicting how a dynamic policy will behave requires the sequence; predicting how a static one will behave does not.

9.3 A fourth confounder: service discipline

§8.3 evaluated affinity batching under a deadline-driven rotation in which up to 24 sessions are admitted and eight are served per step. We had also measured the same request set under standard continuous batching. The two disagree about which residency policy is better.

Table 20. Static versus dynamic residency under two service disciplines. (*Identical 72 requests, 23,529 decode forwards, capacity and pin list; 128-expert model, $\rho = 40\%$.*)

service discipline	mean batch	static pinned	LFRU	static vs LFRU
continuous batching, B = 8	7.69	19.37%	18.01%	−7.5%
deadline rotation, 24 admitted / 8 served, W = 4	7.83	19.15%	23.08%	+17.0%

The static arm barely moves (19.37% $\rightarrow$ 19.15%); LFRU degrades by five percentage points. Rotating a subset of admitted sessions repeatedly disrupts the recency and frequency state on which dynamic policies depend, while a fixed set has no state to disrupt.

The two disciplines are not equally representative. Production continuous batching advances every admitted sequence each iteration up to a concurrency limit; queued requests wait rather than being rotated in, and rotation arises mainly under memory-pressure preemption. The rotation used in §8.3 was chosen to create the scheduling slack affinity selection requires, not because it models a default deployment.

We report this as a measurement result rather than a design recommendation. It means that **a residency-policy comparison that does not state its service discipline is not interpretable**, on the same footing as the three axes of §4–§6. We did not anticipate it, and we found it only because two experiments run for different purposes happened to disagree.

9.4 Gate-mass rank is descriptive, not a quality result

Every mechanism in §8 attempts to reduce the *number* of misses. An orthogonal approach reduces the *bytes per miss*: serve low-weight experts at reduced precision while keeping high-weight experts at full precision. This changes traffic without changing hit rate, so it is untouched by the Belady bound.

Its viability depends on how unevenly gate mass is distributed across the top-*k* ranks. On the 128-expert model it is nearly flat:

rank	1	2	3	4	5	6	7	8
share of gate mass	21.5%	16.4%	13.7%	11.9%	10.5%	9.4%	8.6%	8.0%

The highest-ranked expert carries only $2.7\times$ the mass of the lowest, and ranks 1–4 account for 63.5% of the total. This says that rank is a weak byte-allocation heuristic on this model. It does **not** measure the quality loss from reducing precision: gate mass is not causal output importance, quantization error is not equivalent to dropping a contribution, and no mixed-precision implementation was evaluated. We therefore retain the distribution as a design warning, not as evidence that mixed precision is closed or that a larger model will be flatter.

A cross-model comparison at aligned regimes suggests the static disadvantage may grow with expert count, but the evidence is two comparable points on heterogeneously collected traces; we report it in Appendix B as hypothesis-generating only.

§10 Benchmarking Recommendations

Each of the preceding sections identifies an evaluation choice that is rarely stated and that changes conclusions when it varies. We collect them here as a reporting checklist. Every item is one we failed to report, or reported incorrectly, at some point in this work; the section reference gives the evidence that motivated it. We apply the checklist to ourselves in §13.

The intent is comparability rather than compliance. Most items ask an author to *state* a choice, not to make a particular one. Two items — A2 and D3 — are stronger, because the practices they exclude are not alternative modelling choices but errors.

10.1 Group A — Replay and execution model

Applicability. Group A applies to trace-driven or otherwise modelled cache evaluation. For an end-to-end real-system experiment, the implementation itself defines the transfer unit and retention order; A1 should report that contract, while A2–A4 may be marked not applicable rather than imposed retroactively.

A1. State the execution unit. Is hit/miss classified once per (**step, layer**) event over the batch-wide expert union, or per individual expert access? The two are not interchangeable. (§2.1)

A2. State whether retention decisions are deferred to the event boundary. If they are not, state explicitly whether a block resident at event start and required later can be evicted and counted again. Such a replay is inconsistent with one-transfer-per-distinct-expert fused-event accounting; it may instead describe a genuinely serialized execution, which should be named as a different contract. (§2.2, §4)

A3. If replay is sequential, state the intra-event ordering rule. Ascending expert index makes checkpoint numbering part of the retention decision; randomizing it removes that dependence but not the error in A2. (§4.1–§4.2)

A4. Provide a small hand-verifiable trace or equivalent conformance check, so that a reader can confirm which semantics an implementation realizes. (§2.5)

10.2 Group B — Workload construction

B1. Report the number of distinct instruction templates per workload category, together with the number of distinct prompt prefixes and distinct final lines per category. A single template per category is a risk indicator, not a defect by itself. (§5.6)

B2. State whether the model’s chat template was applied. Under raw continuation the model may complete the instruction rather than answer it, producing shared prefixes across all requests in a category. (§5.2)

B3. Report the number of decode steps recorded and whether the analysis window extends beyond the shared-prefix region. Short windows are dominated by it. (§5.2)

B4. State whether concurrent requests are position-locked — equal length, lock-step admission and retirement — since this aligns any shared prefix at identical scheduling steps and inflates per-step union effects. (§5.2)

B5. Report positional token agreement and the number of unique generated sequences per category. (§5.3)

B6. Report within-category expert overlap restricted to request pairs whose generated text barely overlaps, against a cross-category baseline. The residual is the portion of the effect not explained by surface repetition. (§5.3)

B7. Report whether the effect is stable across decode-position bands. An effect concentrated in the same band as a shared generated prefix should be treated as surface repetition until shown otherwise. Position decay without a shared prefix can be a genuine response-phase effect. (§5.4)

B8. Where feasible, include a matched-pair control in which the same source records are rendered under both constructions, which converts the artifact from an argument into a measurement. (§5.5)

10.3 Group C — Regime and comparability

C1. Report the per-step, per-layer expert-union distribution, the per-layer capacity, event-specific $r(s,1)$, and at least its mean r_bar and fraction above one. Without these a result cannot be placed; the mean alone is not a sufficient statistic. (§6.1, §6.5)

C2. State the cache scope — per-layer or global — and, for per-layer, how the remainder of the budget is allocated. (§2.3)

C3. State the miss denominator: expert assignments before or after intra-event deduplication. (§2.4)

C4. State whether cross-request deduplication is credited to the policy. It is obtained by any batching scheduler and is not a policy contribution. (§2.4)

C5. State the service discipline: continuous batching, rotation of a subset of admitted sessions, admission and preemption rules, and any fairness bound. Changing only this, at fixed request set, capacity and mean batch size, moved a static pinned set from 7.5% worse than LFRU to 17.0% better. (§9.3)

C6. For every static or pinned baseline, state its fit source, split boundary, quota rule, and whether the initial load is counted in reported traffic. A same-trace popularity diagnostic and a discovery-frozen pin list are different baselines even if both are called “static”. (§2.3–§2.4, §9.1)

C7. For cross-model comparison, first align on r_bar , not on batch size, and then report remaining architectural differences. Aligned on batch size, three models in this paper span 36.8 percentage points of gap; aligned on r_bar , the same measurements agree to within 1.1–4.6. (§6.4)

10.4 Group D — Interpreting oracle bounds

D1. Report absolute transferred bytes per output token alongside any gap, together with the bandwidth budget implied by the target service rate. A small gap is ambiguous between “existing policies suffice” and “no policy is viable”, and only the absolute figure distinguishes them. (§6.6)

D2. When motivating a controller by an offline-optimal gap, report the forced-admission decomposition, so that the share of the gap resting on future-victim ranking is visible rather than implied to be available. (§7.2)

D3. Do not report per-step best-of-N batch compositions as achievable scheduling gains. Estimate scheduling headroom on complete trajectories under the intended fairness constraint; the per-step figure in this work overstated the trajectory figure by a factor of roughly three. (§8.4)

D4. State the scope of every negative result: model, residency fraction, concurrency, service discipline, fairness constraint, and whether service is strictly lossless. One attempted negative result in this paper was invalidated by its own capacity floor and would be misleading without this scope. (§8.2, §8.5)

Appendix B gives a minimal reporting block, compact enough to include verbatim in a paper or artifact README, covering the items above that admit a one-line answer.

10.6 What the checklist does not cover

The checklist addresses trace-driven evaluation of residency and scheduling. It says nothing about several things that matter for a deployment claim and that we did not measure: real host-to-device transfer and its overlap with compute, interconnect topology and achievable sustained bandwidth, kernel time, memory contention between expert cache and attention state, and any quality-lossy mechanism. A study that satisfies every item here has established comparability with other trace-driven studies, not deployability.

We also expect the list to be incomplete. §9.3 documents a confounder we did not anticipate and found by accident, after three others had already been identified; we see no reason to believe it is the last.

§11 Limitations

This paper argues that MoE cache results are sensitive to unreported evaluation choices, which obliges us to state our own boundaries plainly.

Everything here is trace-driven simulation. We replay recorded routing decisions against a cache model and count blocks transferred. We do not measure host-to-device transfer, its overlap with compute, sustained bandwidth, interconnect topology, kernel time, graph-capture interactions, or contention between the expert cache and attention state. Miss bursts are reported in block counts, not in time. A policy that transfers fewer blocks here may not be faster in a system, and one that transfers more may be, if its transfers overlap better. No deployment claim follows from any result in this paper; where engineering units appear they are unit conversions applied to the analytical reference frame of §3.2 and are labelled as such.

Model and scale coverage is narrow. The results of §5, §7, §8 and §9 rest on a single 128-expert model, which is additionally 4-bit quantized; routing under quantization may differ from full precision and we did not compare them. The 40- and 64-expert models appear only for cross-model checks. No trace was collected at the 896-expert scale that motivates the application, and §6.4 shows that its regime ratio is reached by our models only at low concurrency — so the setting the motivation comes from is the one our data covers least well. Regime coverage is also bounded below by expert count: the 40-expert model cannot reach $\mathbf{r_bar} < 0.5$ at $\rho = 40\%$ at any concurrency, so conclusions about slack regimes rest on the 128-expert model alone.

Workload and probe construction constrain what the conclusions mean. We evaluate strictly lossless service only; expert substitution, pruning, merging and reduced-precision fallback are excluded by construction, and §9.4 measures the gate-mass distribution that would govern one such mechanism without evaluating any implementation of it. §8.1 closes prefill eviction and admission only — layer-major versus chunk-major prefill execution ordering was never measured and remains the largest lever this paper does not examine. All traces use greedy decoding, which is reproducible but does not sample production generation diversity. The probe set repackages public sources; its category proportions are a design choice and estimate no deployment’s traffic mix, and the six archetypes are not a taxonomy of enterprise workloads. §9.1 fits a static residency set from one calibration trace and measures transfer to one held-out trace; we do not study how quickly such a set becomes stale under workload drift, which is the question a deployment would ask.

Method-specific and statistical limits. The next-use estimator of §7.3 is a linear model over eight features with no architecture search; a negative result from a single estimator is weaker than the decomposition it accompanies, and we report it because its failure is not marginal and because it is what a practitioner would try first. The forced-admission decomposition is arithmetic given a trace and inherits every limitation above. §9.3 compares two service disciplines, not a family, and §8.3 fixes one fairness setting ($W = 4$, 24 admitted sessions), so its 5.9–7.9% oracle headroom is specific to that constraint. Tie-seed variation of order 10^{-5} bounds simulator nondeterminism, not sampling variability; we report no population confidence intervals for quantities estimated from a single trace. Where draw-to-draw spread is quantified (§5.7) it is an estimate of reference variability from seven size-matched draws, and each homogeneous condition is still two draws of twelve requests. Reported ranges apply to one operating point: the 44.18–45.93% range of §7.1 covers thirteen workload compositions at $\rho = 40\%$, $B = 8$, and must not be pooled with the $B = 2$ condition or the residency sweep, whose gaps span 1.95–44.78%. Discovery and confirmatory splits share no source group but were collected under the same model, decoding configuration and length cap. Finally, the audit in §12 evaluates what each work **reports**, not what it does; it establishes that certain results cannot be compared, not that any is incorrect.

The two that matter most. If one limitation were removed we would choose real-hardware validation: every result here is a block count, and the step to service quality passes through overlap, topology and contention we do not model. If a second were removed we would choose a routing trace at the scale the motivation comes from — because §6 establishes that these results are regime-dependent, the measurement that would most change our confidence is not a better policy or a stronger predictor but a trace from that regime.

§12 Related Work and Reporting Audit

12.1 Expert caching and offloading systems

Recent work attacks MoE memory pressure through expert caching, prefetching, mixed-precision fallback, expert substitution, and CPU–GPU co-execution. Mixtral-Offloading (Eliseev and Mazur 2023), MoE-Infinity (Xue et al. 2025), and HOBbit (Tang et al. 2024) implement end-to-end offloading systems under different hardware and precision assumptions. SpecMD (Hoang et al. 2026) provides a PyTorch-hook framework for composing caching, prefetch, miss-handling, and routing policies and proposes Least-Stale eviction. DALI (Zhu et al. 2026) studies residual-based expert prefetching in a KTransformers-based system, while SP-MoE (Chen et al. 2025) combines expert prefetching with speculative decoding. These works primarily ask which mechanisms improve a specified implementation. We ask a complementary question: what must be stated before a trace-derived policy comparison can be interpreted or transferred?

The distinction matters for Axis I. A real execution system determines its own transfer and retention order; our replay counterexample does not retroactively apply to such a system. It applies to a simulator only when the simulator claims the fused-event traffic contract of §2 while permitting a start-resident, not-yet-served expert to be evicted and counted again inside that event.

12.2 Expert prediction is not next-use prediction

Fate (Fang et al. 2025), Pre-gated MoE (Hwang et al. 2024), SiDA-MoE (Du et al. 2024), and MoE-Beyond (Gavhane et al. 2025) exploit predictability in expert activation. Their prediction targets and system contracts differ: several predict which experts a request will select in an upcoming layer or token so that transfer can begin early; Pre-gated MoE changes the architecture and training procedure; MoE-Beyond predicts token/layer expert activations from token embeddings and layer positions and separately evaluates a trace-based cache simulation at batch size one. Section 7 instead predicts a cached block’s next-use distance for victim ranking and bypass admission. A positive result on next-expert prediction is therefore neither a counterexample to nor evidence for our negative next-use result.

12.3 Audit of reported evaluation contracts

We audited the public text of ten representative papers (Appendix A): seven end-to-end offloading systems, one trace-driven cache simulation, and two architecture/prediction studies. We record what each paper reports, not what an unreleased implementation may do. Axis I has narrow applicability — for the seven end-to-end systems the implementation, not a trace replay, defines execution, and the single trace-driven study assumes batch size one, which removes the cross-request union central to our conditions. Beyond that, none of the ten directly reports the measured per-step/per-layer expert union divided by usable per-layer capacity, so the r_{bar} required by §6 cannot be reconstructed without the underlying routes; and while papers identify datasets and often prompt or generation lengths, the number of instruction templates per category, chat-template application and positional synchronization are generally not reported, which prevents applying the diagnostic of §5 from the paper alone. Missing fields do not imply contamination.

This audit does **not** show that any cited speedup or quality result is wrong. Many evaluate targets outside our scope, and real-system measurements avoid the specific replay failure by construction. It shows that the public reports do not expose a common evaluation contract sufficient to compare cache-policy numbers across systems and models. Appendix A gives the per-paper evidence and puts our own invalidated analyses in the first row.

§13 Artifacts and Conclusion

13.1 Released artifacts

Simulator. The event-atomic replay engine of §2, with per-layer and global cache scope, seeded tie-breaking, and the eight policies of §2.3 including the `Belady-forced-admit` decomposition probe of §7.2.

Conformance trace. The hand-verifiable trace of §2.5 — 3 layers, 4 experts per layer, 5 scheduling steps, 30 union accesses, yielding 12 hits and 18 misses under Definition 2 — with the expected counts, so that an independent implementation can confirm which semantics it realizes.

Workload probe set. ControllerProbe-D1: 432 records across six workload archetypes, twelve prompt forms crossed with two instruction languages, eight task framings and five payload rendering styles per archetype, a group-aware discovery/confirmatory split, and the matched-pair fixed-template control arm of §5.5. Built entirely from public benchmark and public industrial sources.

Contamination audit tooling. The prompt-side and route-side diagnostics of §5.3–§5.4, implemented to run on any probe set and any route collection, not only ours. This is the component we expect to be most directly reusable.

Routing traces and event streams. Where upstream licences permit redistribution, decode routing for three models, including the corrected collection of §3.3 and the earlier single-template collection retained as the contaminated condition. Otherwise the release contains collection/conversion scripts, frozen selections, manifests and cryptographic hashes rather than silently redistributing restricted source material.

Pre-registration documents, unchanged. Every frozen criterion with the split it was to be evaluated on, including those in which a criterion failed and the associated development line was stopped, and including the design revision recorded in §8.6.

Derived result artifacts, each with per-condition rows and input manifest hashes, notably the reproducible v2 thirteen-condition gap-invariance measurement of §7.1, the regime ablation of §6.5, the natural-C4 false-positive check, its labelled 64-token synthetic-prefix positive control, and the static protocol audit that freezes the evaluated event-manifest hash, discovery pin hash, quota rule, preload convention, and both static values used in the text.

We apply the checklist of §10 to this paper in Appendix B, including one item we cannot complete: we set no service-rate target for the primary model, so the bandwidth budget required by D1 is unavailable without the hardware calibration §11 says we lack. The invalidated analyses recorded during this work are listed in Appendix A.

13.3 Conclusion

We set out to measure how much room a workload-aware expert cache controller has in MoE inference. The measurement turned out to be the harder problem.

Three evaluation choices — how a fused-event traffic contract is replayed, how a workload-partitioned probe set is constructed, and what operating regime a result sits in — each change conclusions rather than merely shift numbers. The first inverts policy rankings by selectively penalizing recency-based policies. The second manufactures apparent workload locality large enough to reverse which workloads look cache-friendly. The third makes cross-model comparison uninterpretable unless the ratio of per-step expert union to cache capacity is reported, and even then does not license extrapolation. A fourth — the service discipline — we found only by accident, after the first three were already identified.

With all of them controlled, a large and stable gap to the offline optimum remains, and it is mostly not what it appears to be: 84.3–96.6% of it rests on knowing which resident block is used furthest in the future, and the natural online estimator of that quantity performs worse than the baseline it replaces. Three natural mechanisms failed to reach pre-registered thresholds in the settings we measured; a fourth experiment, semantic partitioning, was itself invalidated because its dedicated partitions could not hold one request’s top-k set. A fixed, offline-chosen residency set remains a useful transparent reference whose union-denominator traffic is determined by an access-count distribution rather than access order. It never wins under standard continuous batching, is close to LFRU only at high local concurrency, and is substantially worse in the slack regimes most relevant to our unmeasured frontier-scale motivation; it is not a validated product mechanism.

We state our position narrowly. We have not shown that expert caching is a closed problem, that future-victim distance is unpredictable in principle, or that online controllers have no future. We have shown that **in our evaluated settings a large offline-optimal gap substantially overstates the gains recovered**

by **representative lightweight causal mechanisms**, and that reporting such a gap without decomposing it invites an inference the evidence does not support.

The portable output of this work is not any individual number — §6 establishes that the numbers are regime-bound — but the reporting checklist of §10, the conformance trace that pins down the replay semantics, the contamination diagnostics that need no control arm, and the released artifacts. We expect the checklist to be incomplete, and we would consider a fifth confounder found by a reader to be a use of it rather than a refutation.

Appendix A — Related-work reporting audit

Date audited: 2026-08-02.

Rule: **not reported** means only that the public paper text did not expose the field. It is not an assertion about private code or actual execution.

The audit freezes the following public versions: SpecMD v1, DALI v1, Fate v2, Pre-gated MoE v3, SiDA-MoE v2, MoE-Beyond v1, Mixtral-Offloading v1, HOBbit v2, MoE-Infinity v3, and SP-MoE v2. Titles and version identifiers were rechecked against the official arXiv abstract pages on 2026-08-02; detailed evaluation claims were checked against the corresponding PDFs. `paper/references.bib` contains the formal records. A later paper revision must be re-audited rather than silently substituted.

A.1 Self-audit first

item	original state	correction	status
offline oracle	forced every miss to be admitted	implemented bypass-capable Belady; retained forced-admit only as a decomposition probe	superseded
replay	flattened an event in ascending expert-ID order	event-start hit/miss classification and boundary retention	conclusion about admission/pinning withdrawn
workload probe	one category template, raw continuation, 63 decode steps	chat template, diverse wrappers, 384-step cap, natural EOS, staggered arrivals, matched control	industrial-locality headline withdrawn
scheduling oracle	per-step best of 300 random batches: 19.3%	complete trajectories, W=4: 5.9–7.9%	old headroom estimate withdrawn
semantic partitions	best dedicated partition had 6 slots for top-8 routing	identified that the grid could not hold one request’s single-step set	treated as design failure, not mechanism refutation
condition count	verbal claim of 17; first 13-condition artifact lacked frozen selections	deterministic 13-condition v2 with IDs, offsets, condition hashes and source hash	v1 superseded

A.2 Per-paper extraction

Abbreviations: **A** end-to-end system; **B** trace-driven simulation; **C** architecture/training co-design; **D** prediction study. “Union ratio” means the measured per-step/per-layer union divided by usable cache capacity, not a null model estimate.

Table A1. Per-paper extraction from the frozen public versions. “NR” means not reported in the public paper.

work	class and evaluation vehicle	reported experimental setting	reporting interpretation
SpecMD	A; PyTorch hooks on one A100-80GB with software capacity/bandwidth constraints	GSM8K, TruthfulQA and NaturalQuestions; autoregressive generation; 1/5/25% capacity; 5GB/s example; active batch and chat formatting NR	End-to-end path, so event replay is N/A; union ratio and template multiplicity NR. Comparable policy names do not imply the same execution contract or denominator.
DALI	A; KTransformers-based RTX 3090 system	C4 with WikiText calibration; default prompt/output 64; batch-size sweeps; often 50% cache with some 25% ablations; chat formatting and template multiplicity NR	Event replay is N/A and union ratio is NR. DALI reports real throughput; Axis I is not a criticism of it.
Fate	A/D; two-PC implementation	ChatGPT-prompts, HumanEval and GSM8K; decode output up to 1024; prefill lengths 128/256/512; memory budgets reported; active decode concurrency, chat formatting and template multiplicity NR	Event replay is N/A and union ratio is NR. Fate predicts next experts for prefetch, not next-use distance.
Pre-gated MoE	C; trained architecture and FasterTransformer implementation	XSum, closed-book QA and SQuAD; primary system focus at batch 1; model/cache design and task datasets reported; chat template not applicable or NR	Event replay is not the paper’s question and union ratio is NR. The method modifies training and the routing contract, outside strictly post-hoc caching.
SiDA-MoE	C/D; hash-based predictor/system design	C4 and Switch-family evaluation; model configuration reported; category-template risk not applicable to the reported corpus setup	Event replay is not the primary question and union ratio is NR. Its prediction objective differs from victim next-use ranking.
MoE-Beyond v1	B/D; token/layer activation predictor plus trace-based simulator	6,994 Puffin training prompts and 100 WebGLM-QA test prompts; batch size 1; cache-capacity sweep; chat formatting NR	Token-by-token replay is described, but fused-event retention details and union ratio are NR. Batch 1 removes cross-request union, leaving insufficient detail for an event-contract comparison.

work	class and evaluation vehicle	reported experimental setting	reporting interpretation
Mixtral-Offloading	A; end-to-end offloading implementation	OpenAssistant conversations with autoregressive sampling; practical focus at batch 1; per-layer cache sizes and source conversations reported; chat-template details NR	Event replay is N/A and union ratio is NR. Its real-system latency is not invalidated by our replay result.
HOBBIT	A; llama.cpp-based CPU-GPU system	Alpaca for performance; GSM8K and TruthfulQA for quality; several input/output lengths; batch 1; cache/precision budgets reported; chat formatting and template multiplicity NR	Event replay is N/A and union ratio is NR. Lossy precision fallback is outside our strictly lossless scope.
MoE-Infinity	A; RTX A5000 end-to-end system	290 BIG-Bench/FLAN/MMLU tasks; prompt 512/output 32 in the main latency test; local deployment, effectively batch 1; task rendering insufficient for §5 diagnostics	Event replay is N/A and union ratio is NR. Request-level activation maps and real execution answer a different question.
SP-MoE	A; speculative decoding and expert-prefetch system	Four public datasets; autoregressive evaluation; batch size 1; chat/template multiplicity NR	Event replay is N/A and union ratio is NR. Its prefetch/speculation target differs from cache-victim prediction.

A.3 What may and may not be concluded

The audit supports three reporting claims: most audited systems do not expose a directly comparable $\mathbf{r_bar}$; prompt-surface contamination cannot generally be checked from their papers; and our replay-semantics warning should not be aimed at end-to-end implementations. It does not support re-labelling published speedups as artifacts, assigning a direction to an unreported implementation choice, or inferring that a dataset used one fixed prompt merely because prompt construction was omitted.

Appendix B — Supplementary material

Material referenced from the body but not required to follow the argument.

B.1 A minimal reporting block

The following is compact enough to include verbatim in a paper or artifact README, and covers the items above that admit a one-line answer.

```
Execution model
  event unit ..... (step, layer) union | per-access
```

```

retention decision ..... event boundary | immediate
intra-event order (if immediate) ... ascending id | randomized | n/a
conformance trace ..... yes | no

```

Workload

```

templates per category ..... N
distinct prompt heads / finals ..... N / N
chat template applied ..... yes | no
decode steps recorded ..... N
position-locked cohorts ..... yes | no
unique generated sequences ..... N / N
positional token agreement ..... mean, p95
expert Jaccard | low-text-overlap .. value (cross-category baseline: value)
effect stable across position bands yes | no | n/a

```

Regime

```

experts N, top-k, layers ..... N / k / L
residency fraction rho ..... value
cache scope ..... per-layer | global
union per layer-step ..... mean, p95
capacity per layer ..... value
r(s,l) = union / capacity ..... mean, p95, fraction > 1
concurrency ..... B (and admitted sessions if rotating)
service discipline ..... continuous | rotation(W=..) | other
miss denominator ..... pre-dedup | post-dedup
dedup credited to policy ..... yes | no
static fit source ..... same trace | discovery split | n/a
static quota rule ..... description | n/a
initial pinned load counted ..... yes | no | n/a

```

Oracle

```

offline bound reported ..... yes | no
forced-admission decomposition ..... admission % / future-victim %
absolute bytes per output token .... value
bandwidth budget assumed ..... value (and target service rate)

```

B.2 The checklist applied to this paper

We fill the reporting block of §10.5 for the primary configuration. One field we cannot complete, and we say so rather than omitting it.

Execution model

```

event unit ..... (step, layer) union
retention decision ..... event boundary
intra-event order ..... n/a
conformance trace ..... yes (§2.5)

```

Workload

```

templates per category ..... 24 form x language (diverse arm)
                                     1 (matched control arm, by design)
distinct prompt heads / finals ..... 23-24 / 16-22 per 24-record cell
chat template applied ..... yes, reasoning mode disabled
decode steps recorded ..... up to 384, natural EOS
position-locked cohorts ..... no (arrival offsets 0-23)
unique generated sequences ..... 12 / 12 per archetype sample

```

```
positional token agreement ..... 0.45-1.06% mean
expert Jaccard | low-text-overlap .. 0.085-0.144 (cross-category: 0.077)
effect stable across position bands yes for 2 of 6 archetypes (§5.4, §5.7)
```

Regime

```
experts N, top-k, layers ..... 128 / 8 / 48
residency fraction rho ..... 0.40
cache scope ..... per-layer, remainder to lowest indices
union per layer-step ..... 42.63 mean, 50 p95
capacity per layer ..... 51-52
r(s,l) = union / capacity ..... mean 0.833; fraction >1: 2.19%
concurrency ..... B = 8
service discipline ..... continuous batching (FCFS);
rotation variant reported separately
miss denominator ..... pre-dedup (logical assignments)
dedup credited to policy ..... no
static fit source ..... discovery split (Static-frozen);
evaluation trace (diagnostic only)
static quota rule ..... equal per layer; remainder to lowest indices
initial pinned load counted ..... yes
```

Oracle

```
offline bound reported ..... yes
forced-admission decomposition ..... 15.69% / 84.31% (B=8)
3.40% / 96.60% (B=2)
absolute transfer ..... 69.2 blocks per output token
(best causal, held-out)
bandwidth budget assumed ..... NOT APPLICABLE - no service-rate target
is assumed for the primary model
```

The last field is a gap in our own reporting. Item D1 asks that a gap be accompanied by the bandwidth budget implied by a target service rate, and for the primary model we set no such target: the measurements are in blocks, and converting them to a budget would require the hardware calibration §11.1 says we do not have. We record the block count instead, and note that a reader wishing to apply D1 to our numbers must supply the missing conversion themselves.

For completeness, the failures recorded during this work and released with the pre-registrations are: an initial offline bound that forced admission of every miss and was therefore not the offline optimum; the sequential replay of §4, whose correction withdrew a conclusion about admission and pinning; the single-template probe set of §5, whose correction withdrew a confirmatory conclusion about industrial workload locality; a scheduling headroom estimate of 19.3% that fell to 5.9–7.9% when measured on complete trajectories (§8.4); a category partitioning experiment that could not have produced a positive result (§8.2); and, during internal review, an unsourced condition count that was corrected from seventeen to thirteen. The first 13-condition artifact still lacked frozen request selections and was superseded by a deterministic v2 with every request ID, arrival offset, condition hash and source-manifest hash.

B.3 Does the static disadvantage grow with expert count?

Referenced from §9. **Hypothesis-generating only:** two comparable points on the expert-count axis, heterogeneous models, and all four rows collected under the single-template protocol of §5, whose surface repetition inflates temporal locality and therefore plausibly inflates the static disadvantage in every row. The comparison is internally consistent; the levels are not trustworthy, and a two-point extrapolation across a further factor of seven in expert count would not be.

Table 18 measures one model. Comparing across models requires aligning on the regime variable of §6, and doing so suggests a trend we can report but not establish.

Table B1. Static versus dynamic at aligned operating regimes. ($\rho = 40\%$, per-layer scope. *Pin lists fitted on the evaluation trace itself — deliberately leaky, so the static arm is shown at its best case and the trend is conservative.*)

model	experts	$\mathbf{r_bar}$	static	LFRU	static vs LFRU
40-expert	40	0.500	31.15%	25.91%	−20.2%
64-expert	64	0.312	35.64%	25.65%	− 39.0%
128-expert	128	0.293	19.57%	12.83%	− 52.6%
128-expert	128	0.156	19.87%	10.57%	−87.9%

Two directions are visible: at comparable $\mathbf{r_bar}$, the disadvantage grows with expert count (−39.0% at 64 experts, −52.6% at 128); and within a single model it grows as the regime becomes slacker (−52.6% at $\mathbf{r_bar} = 0.29$, −87.9% at $\mathbf{r_bar} = 0.16$). Both are consistent with the mechanism of §9.1 — more experts means a flatter popularity distribution, so a fixed top-fraction captures less of the traffic, while a dynamic policy still tracks a working set that remains small relative to capacity.

We label this hypothesis-generating rather than a result. There are only two comparable points on the expert-count axis; the models differ in training and tokenizer as well as in expert count; and all four rows use the single-template collection of §5, whose surface repetition inflates temporal locality and therefore plausibly inflates the static disadvantage in every row. The comparison is internally consistent, but the levels are not trustworthy and a two-point extrapolation across a further factor of seven in expert count would not be.

B.4 The invalidated partition grid

Referenced from §8.2. Retained so that the design failure is inspectable, not as a negative result for semantic partitioning.

The partitioning experiment gave a dedicated per-layer slot budget to each of the two archetypes that §5.7 identified as genuinely cache-friendly, with the remaining four sharing a third partition. Total capacity was held exactly equal to the shared baseline and the remainder was assigned to the shared partition, so the dedicated partitions received no arithmetic advantage. A seven-point allocation grid was swept on the calibration split, the best allocation frozen, and the held-out split evaluated once. Every allocation lost, and the frozen one increased transfer by 115.9%.

The natural reading is that semantic partitioning destroys cross-category expert sharing and duplicates commonly used experts across partitions. We tested that reading and it is largely wrong. Running the *shared* cache at each partition’s own slot count gives 61.08% miss at 6 slots and 27.25% at 39 slots, against 18.76% at the full 51. Weighting these by each partition’s share of logical expert assignments (0.1415, 0.1557, 0.7028) predicts

$$0.1415 \times 61.08\% + 0.1557 \times 61.08\% + 0.7028 \times 27.25\% = \mathbf{37.30\%}$$

against a measured 37.99%. **Capacity fragmentation alone accounts for 96% of the degradation;** sharing and duplication together account for the remaining 4%.

The root cause is arithmetic. Top-k is 8, so a single request touches eight experts at every layer, while a dedicated partition in the best allocation holds six. No partition in the grid could retain even one request’s per-step working set, and the grid’s largest dedicated allocation, 16 slots, holds two. **The experiment as designed could not have produced a positive result**, and we say so rather than presenting the outcome as a mechanism-level refutation.

Whether the arithmetic generalizes is a separate question, and for the frontier-scale reference frame of §3.2 it does not: 358 per-layer slots divided six ways gives approximately 59.7 per partition against a per-category per-step union of roughly 16–26, a margin of 2.3–3.7 $\times$. Hard partitioning may still lose there for the sharing and duplication reasons, but it would not be excluded by capacity arithmetic, and the −115.9% figure must not be carried across.

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